

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #5

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Satellites as Spacecraft (s/c): Topics (Chapter 3)

- Communications satellites (last week)
- AOCS (last week)
- TTC&M
- Power system
- Communications system
- Antennas
- Reliability and Space Qualification

TTC&M

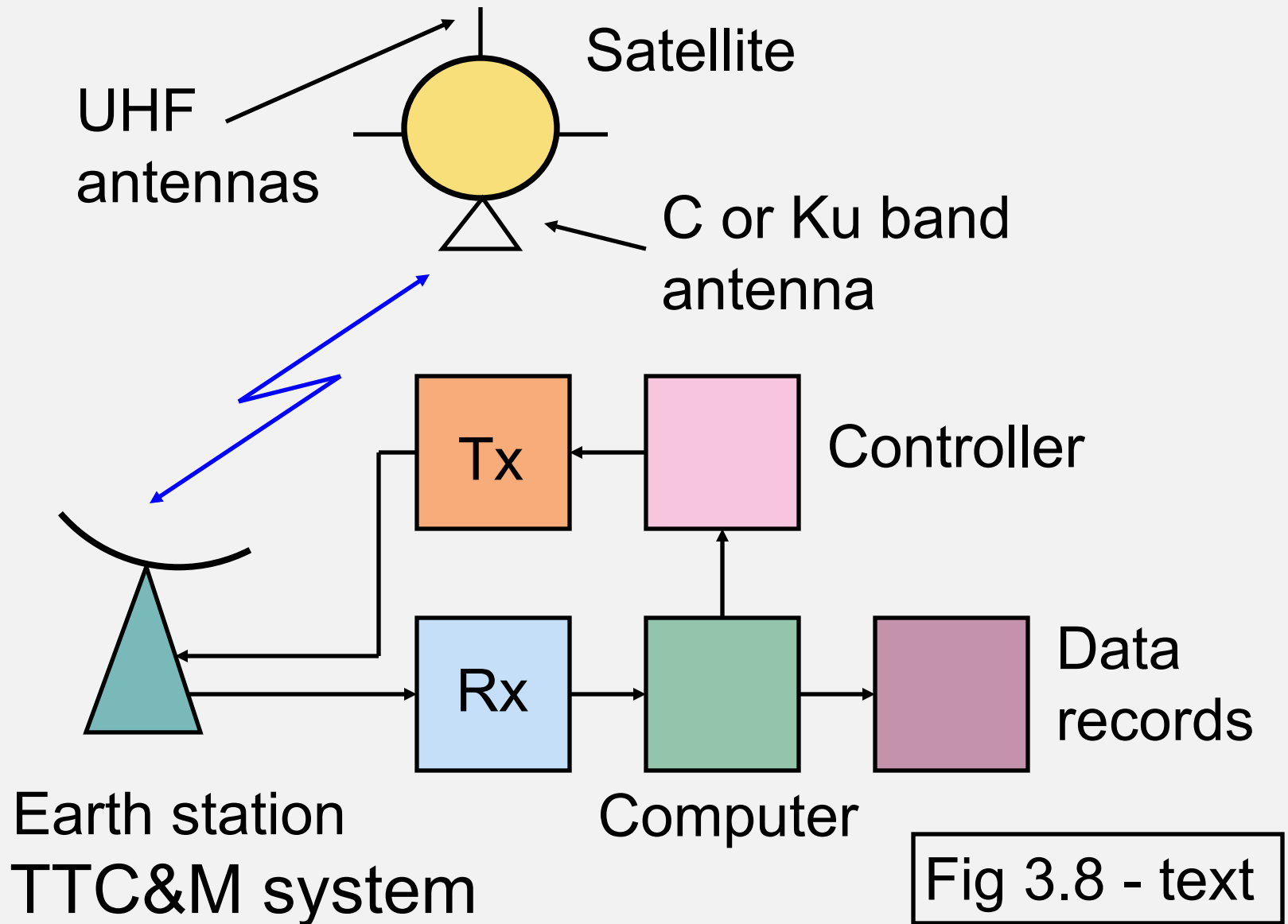
- A typical GEO satellite costs \$250 M on orbit
- Must be monitored and controlled at all times
- Loss of control = loss of \$250 M investment
- Dedicated TTC&M station monitors satellite continuously
- Telemetry system reports thousands of sensor outputs – voltage, current, fuel, temperature, switch settings, etc

TTC&M

- *Telecommand system* uses encryption
- Carefully controlled sequence to minimize errors
 - request is sent to satellite to make change
 - satellite responds with repeat of message
 - control station issues execute command
 - satellite executes change
 - telemetry system reports new state

TTC&M

- TTC&M system is usually duplicated
- Baseline system uses UHF, omni-directional antennas, large system margin (e.g. 30 dB)
- Can command satellite in any orientation, while tumbling, at any range
- Once satellite is stabilized and deployed TTC&M can switch to main communication system in C or Ku band (narrow beam)



Tracking

- Tracking is used to determine satellite orbit
- Based on measurements of distance to satellite – called *ranging*
- *Ranging tones* are transmitted through satellite transponder
- Comparison of phase of different frequencies leads to precise determination of distance
- Observation of range variation with time yields orbital elements – *ephemeris data*

Power Systems

- Communication satellites derive electrical power from sun's energy
- Above Earth's atmosphere, incident solar power is 1.39 kW / m² (normal incidence)
- Solar cells have conversion efficiency of 20 - 35%
- To generate 10 kW with 22% efficiency cells:

$$A = 10 / (1.39 \times 0.22) = 32.7 \text{ m}^2$$

Solar Cells

- Output of solar cells is reduced when temperature increases
- Output falls by 10 – 15% of initial power output during lifetime of satellite
 - some solar cells fail
 - glass cover gets etched by micrometeorites
 - cells age

Solar Cells

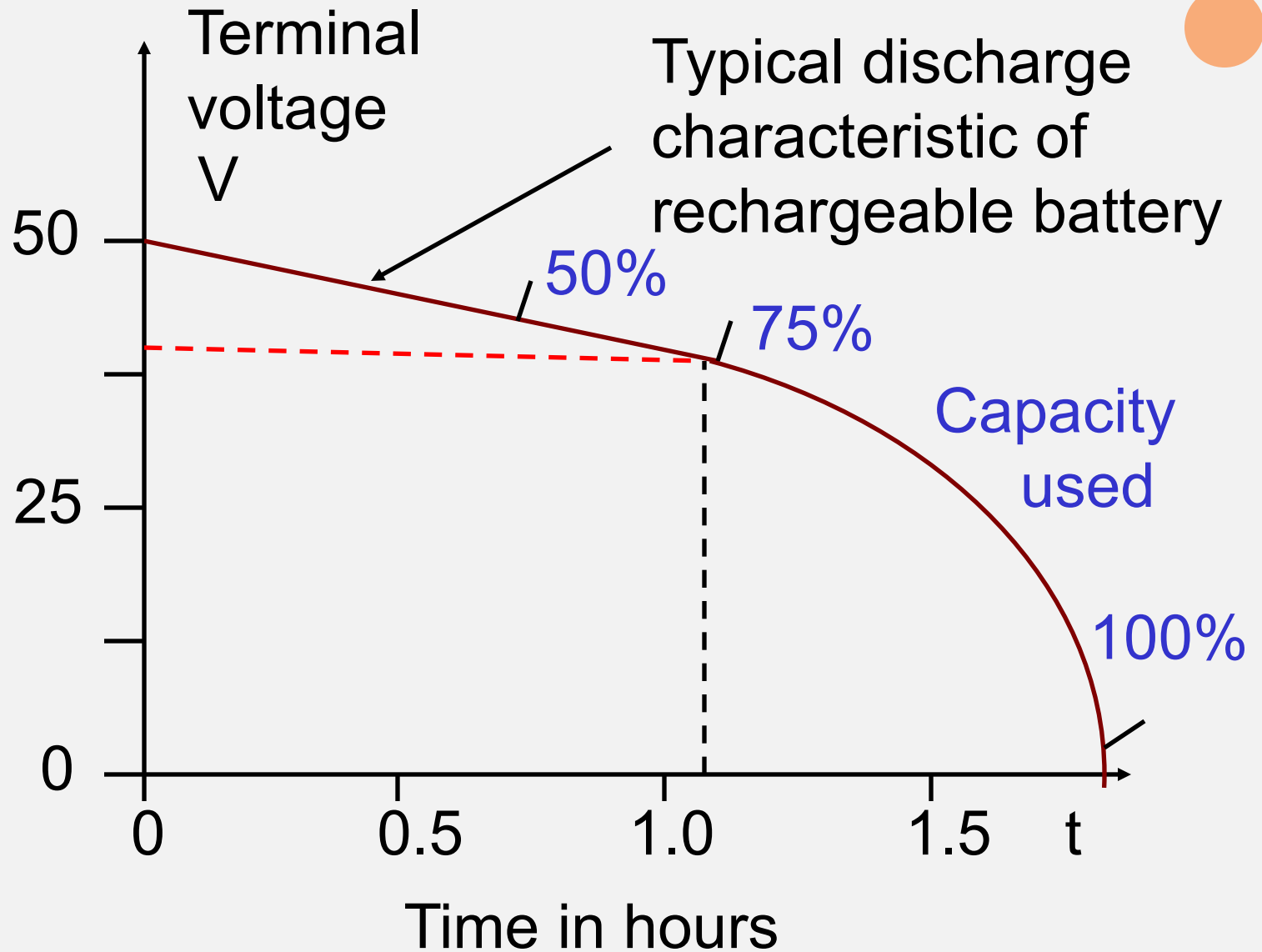
- Output of power system is rated for End of Life (EoL)
- System generates excess power at Beginning of Life (BoL)
- Excess power can be radiated from load resistors facing *cold space*
- Batteries are required for eclipse operation

Batteries

- GEO Satellites need batteries for two reasons:
 - 1) To run vital electronics after launch before solar sails are deployed
 - 2) To provide power during eclipse
- Eclipse example: 5 kW satellite
 - maximum duration of eclipse is 70 minutes
 - batteries must supply 5 kW for 70 minutes
= 5.83 kWhr

Battery Example

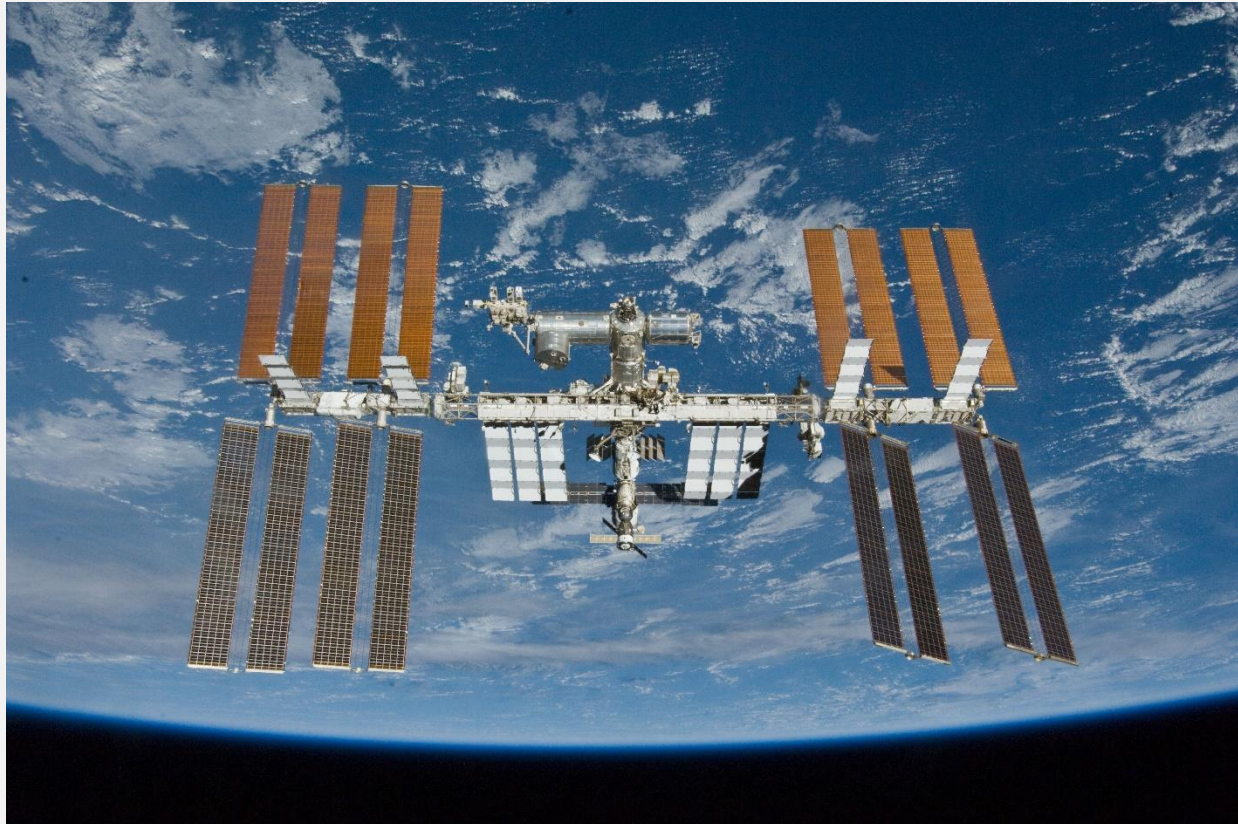
- Typical satellite bus voltage is 50 V
- 5 kW at 50 V requires 100 A
- Available battery capacity required (~ 70 min): 116.7 AH
- Batteries cannot be fully discharged (to $V = 0$ volts)
- Typical discharge is 70%, so battery capacity must be $116.7 / 0.7 = 166.7$ AH



Solar Cells

- A large GEO satellite generates ~ 20 kW of solar panel
- Batteries are required to operate in darkness during eclipse seasons for a small fraction of the orbit (≤ 70 min)
- LEO satellites by contrast must operate on batteries for half of each orbit
- A Starlink satellite is estimated to generate ~ 5 kW of power

International Space Station (ISS)

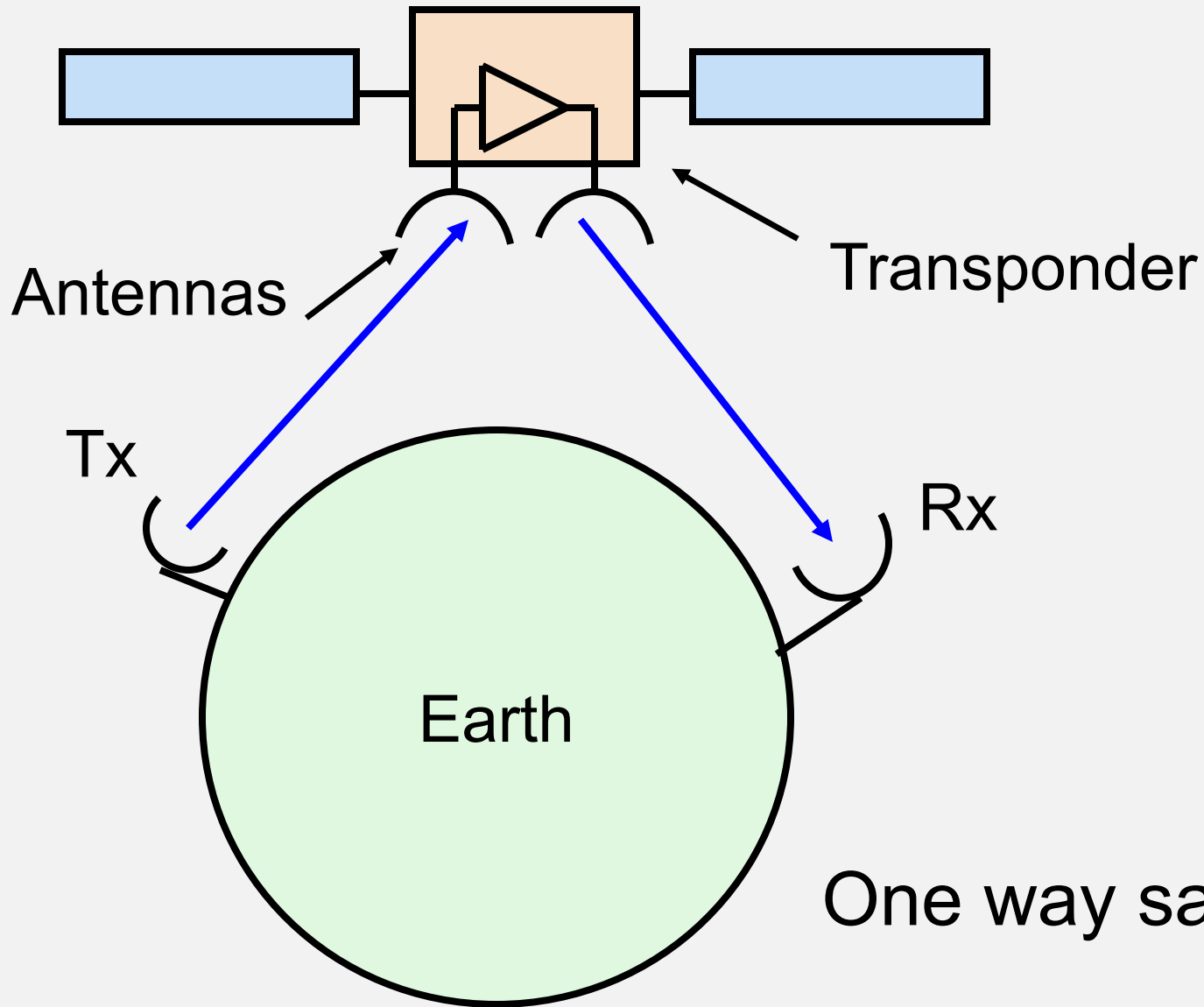


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Solar panels generate 120 KW of power, 60% of output is used to charge batteries

Communications System

- The communications system on a GEO satellite is the payload – earns revenue
- Revenue must pay for satellite, launch, insurance, earth stations, operations
- A typical system of two GEO satellites and one spare costs about \$500M to establish and \$25M per year to operate
- Revenue must cover these costs, pay interest, and make a profit



One way satellite link

Communication System

- The communication system consists of two main parts: antennas and transponders
- Antennas will be covered in another sub-section
- A transponder is a combination of a receiver and a transmitter (also *relay*, *transceiver*)
- A *bent pipe* transponder simply receives a signal from earth, amplifies it, and retransmits back to earth (at a different frequency)

Transponders

- Satellites carry many transponders
- Transponder relays signals back to earth
 - Signal is received, amplified in low noise amplifier
 - Signal frequency is changed, signal is amplified, retransmitted
 - Transmit power typically 40 – 100 W
 - Received signal is weak, typically 10^{-12} W

Frequency Bands

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• HF	3 – 30 MHz	
• VHF	30 – 300 MHz	(10 - 1 m)
• UHF	300 – 1,000 MHz	
• L-band	1 – 2 GHz	(30 - 15 cm)
• S-band	2 – 4 GHz	
• C-band	4 – 8 GHz	(~5 cm)
• X-band	8 – 12 GHz	
• Ku-band	12 – 18 GHz	(~2 cm)
• K-band	18 – 27 GHz	
• Ka-band	27 – 40 GHz	(~1 cm)
• V-band	40 – 75 GHz	(~0.5 cm)

Satellite Frequency Bands

- ITU-R allocates radio frequencies
- World Radio Conference held every few years
- Major Bands: Fixed satellite service (FSS)

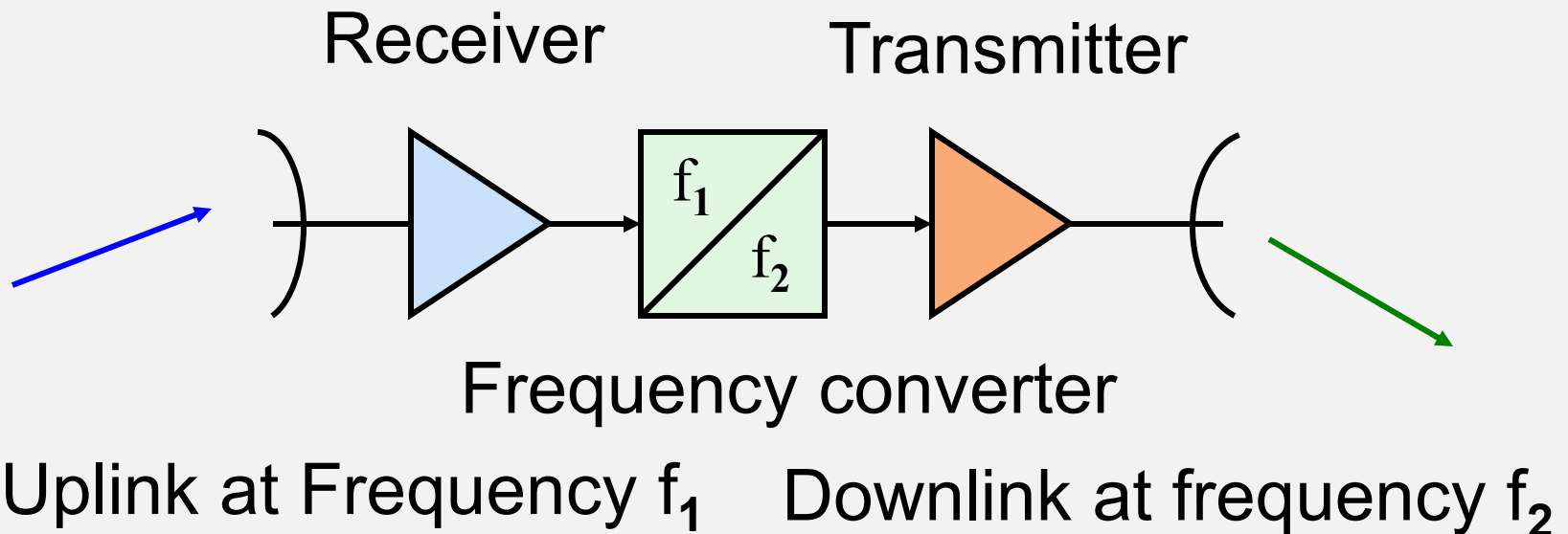
C band: uplink 6 GHz downlink 4 GHz

Ku band: uplink 14 GHz downlink 11 GHz

Ka band: uplink 30 GHz downlink 20 GHz

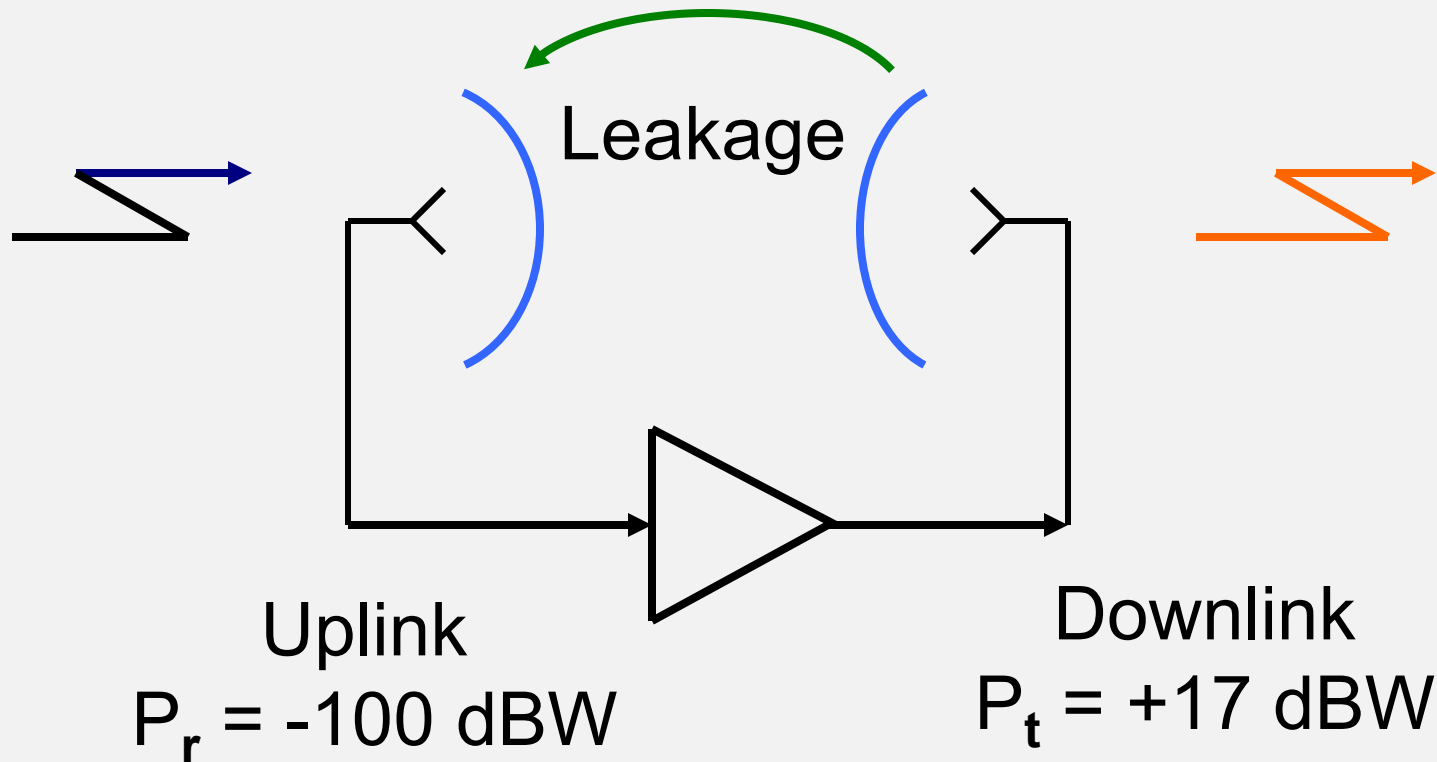
Transponders

- Basic transponder



Transponder

- We have to use a different frequency for receiving (uplink) and transmitting (downlink)
- Using the same frequency would cause amplifiers to oscillate due to feedback
- Transponder gain is 80 – 130 dB
- Transponder requires frequency conversion
- Use amplifiers at several frequencies



Repeater

$$G = P_t - P_r = 117 \text{ dB}$$

Receive and Transmit at same frequency
is not possible – leakage causes feedback

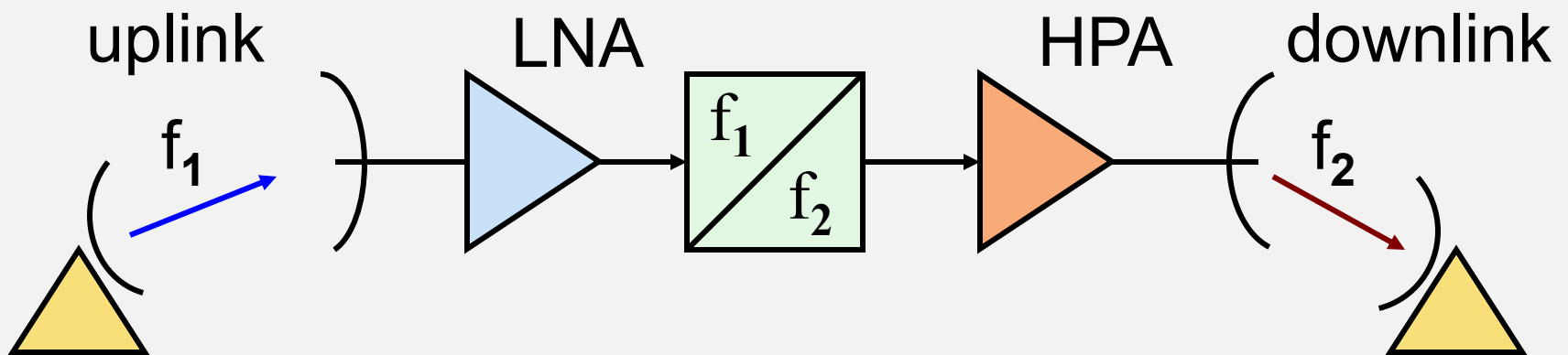
C-Band Satellite Link (6/4 GHz band)

Uplink in near 6 GHz, downlink near 4 GHz

Single conversion with LO = 2225 MHz

E.g. Satellite receives at $f_{in} = 6000$ MHz

Transmits at $f_{out} = 6000 - 2225 = 3775$ MHz



C-band bent pipe Transponder

- Main uplink frequency band is 5.925 GHz to 6.425 GHz (= 500 MHz satellite bandwidth)
- Extension to 5.725 – 7.075 GHz (= 1350 MHz satellite bandwidth)
- Band is divided into *channels*
- Typical channel bandwidth at C band: 36 or 72 MHz
- Assign one transponder to each channel

C-band bent pipe Transponder

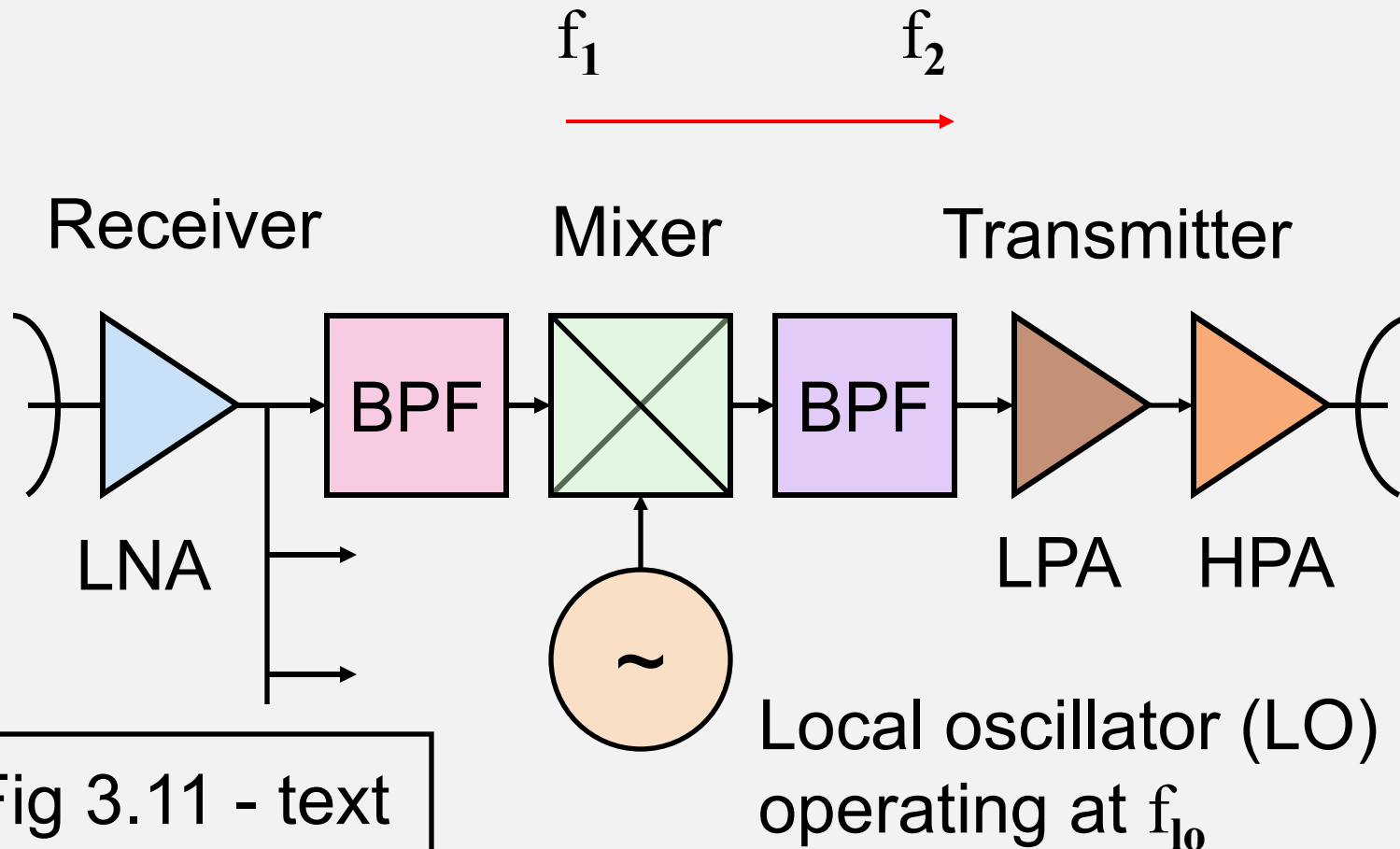


Fig 3.11 - text

C-band bent pipe Transponder

- Receiver leads with a low noise amplifier (LNA)
- The first band pass filter (BPF) selects for the frequency channel assigned to the transponder
- The frequency conversion stage drops the input frequency by 2225 MHz (= LO frequency)
- Two stages of amplification (LPA, HPA) boost signal power to ~ 50 W for transmission

Frequency Conversion

- Frequency conversion requires:
 - Multiplier (mixer)
 - Local oscillator
 - Band pass filter
- Multiplier is actually a switch
- Input / Output frequency is f_{in} / f_{out} where
$$f_{out} = f_{in} - f_{lo} \quad (\text{difference frequency})$$
or
$$f_{out} = f_{in} + f_{lo} \quad (\text{sum frequency})$$
- And f_{lo} is the *Local Oscillator* (LO) frequency

Frequency Conversion

- We can convert a signal to a different frequency by multiplying the signal with a sine wave at frequency f_{lo} , ω_{lo}
- Required sine wave is generated by the local oscillator (LO)

Signal:
$$v_s(t) = V_s \cos \omega_s t$$

Local oscillator:
$$v_{lo}(t) = \cos \omega_{lo} t$$

- Multiply:
$$v_o(t) = V_s \cos \omega_s t \times \cos \omega_{lo} t$$

Frequency Conversion

- Trig identity:

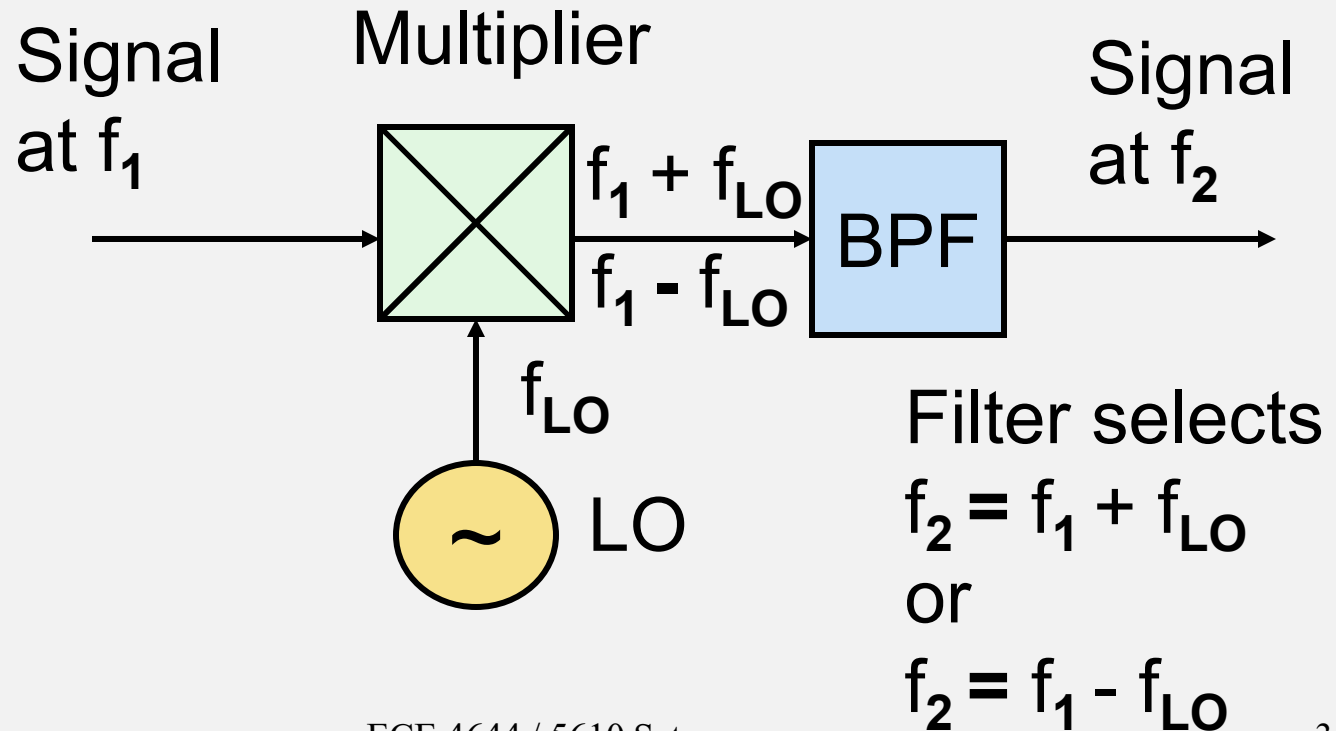
$$\cos A \cos B = \frac{1}{2} [\cos (A + B) + \cos (A - B)]$$

- For signal and local oscillator

$$v_o(t) = \frac{1}{2} V_s \cos (\omega_s + \omega_{lo}) t \\ + \frac{1}{2} V_s \cos (\omega_s - \omega_{lo}) t$$

- $\frac{1}{2} V_s \cos (\omega_s + \omega_{lo}) t$ is the sum *frequency*
- $\frac{1}{2} V_s \cos (\omega_s - \omega_{lo}) t$ is the difference *frequency*
- We use *filters* to select specific bands of frequencies

Frequency Conversion



Frequency Conversion

- Example:

Input : $f_{in} = 5925 \text{ MHz}$

$\times f_{lo} = 2225 \text{ MHz}$

$\rightarrow f_{out} = 3700 \text{ MHz}$

or $\rightarrow f_{out} = 8150 \text{ MHz}$

- A review of frequency conversion theory using a switching multiplier is provided in the Canvas Files area

Mixer Power Levels

- Mixer output power is usually 1 to 5 mW
- Output of transponder HPA is 20 – 100 W
- HPA must amplify signal from mixer
- Gain required is $10 \log (P_{\text{out}} / P_{\text{in}})$
- For 100 W output with 1 mW input

$$G_{\text{max}} = 10 \log (100 / 0.001) = 50 \text{ dB}$$

- Note: all dB calculations are $10 \log (P_2 / P_1)$
- **Never** $20 \log (V_2 / V_1)$

Transponder gain

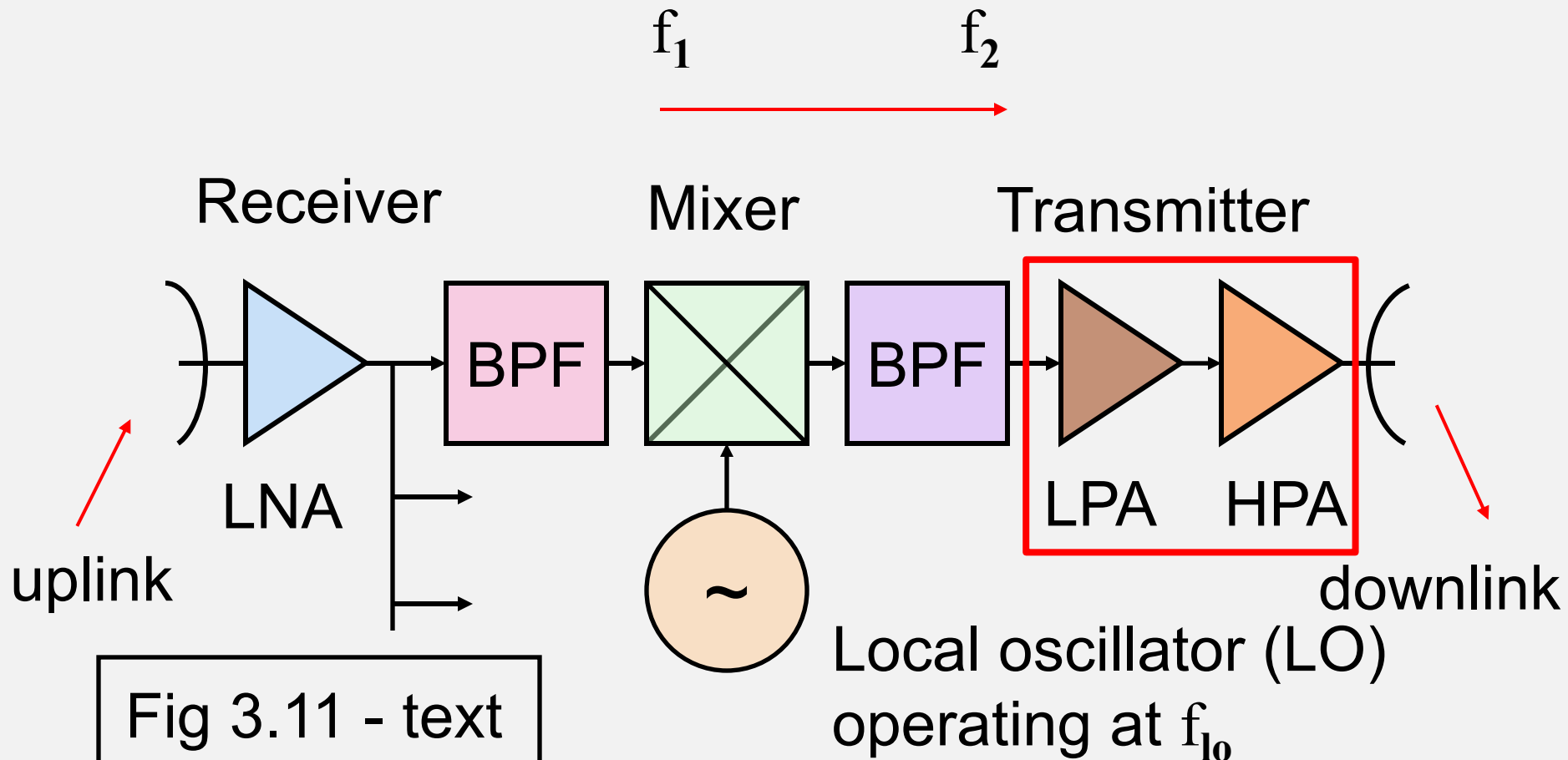
- Typical transponder input power is -100 dBW (10^{-10} W)
- Output of transponder = 100 W = 20 dBW
- Overall gain required is $10 \log (P_{\text{out}} / P_{\text{in}})$

$$\begin{aligned} G_{\text{max}} &= 10 \log (100 / 10^{-10}) = 120 \text{ dB} \\ &= +20 \text{ dBW} - (-100 \text{ dBW}) = 120 \text{ dB} \end{aligned}$$

Transponder gain

- The transponder will also have losses
- Amplifiers must compensate for the losses and achieve ~ 120 dB gain besides
- At microwave frequencies asking for $G > 60$ dB will cause an amplifier to oscillate
- Avoid this by spreading the gain between stages of amplification
- A Low-Power Amplifier (LPA) at the output of the mixer stage might look into a High-Power Amplifier (HPA) that feeds the transmit antenna

C-band bent pipe Transponder



Double Conversion Transponder

- At Ku and Ka band frequencies, transponders use two stages of frequency conversion
- Input signal, f_{signal} , is converted to an IF or *Intermediate Frequency*, f_{IF} , and then to f_{out}
- Transponder IF is typically around 1 GHz
- Filtering is easier to do at 1 GHz than at 11 GHz
- In general, filters should not have bandwidth less than 2% of center frequency ($Q < 50$)

Double Frequency Conversion Transponder

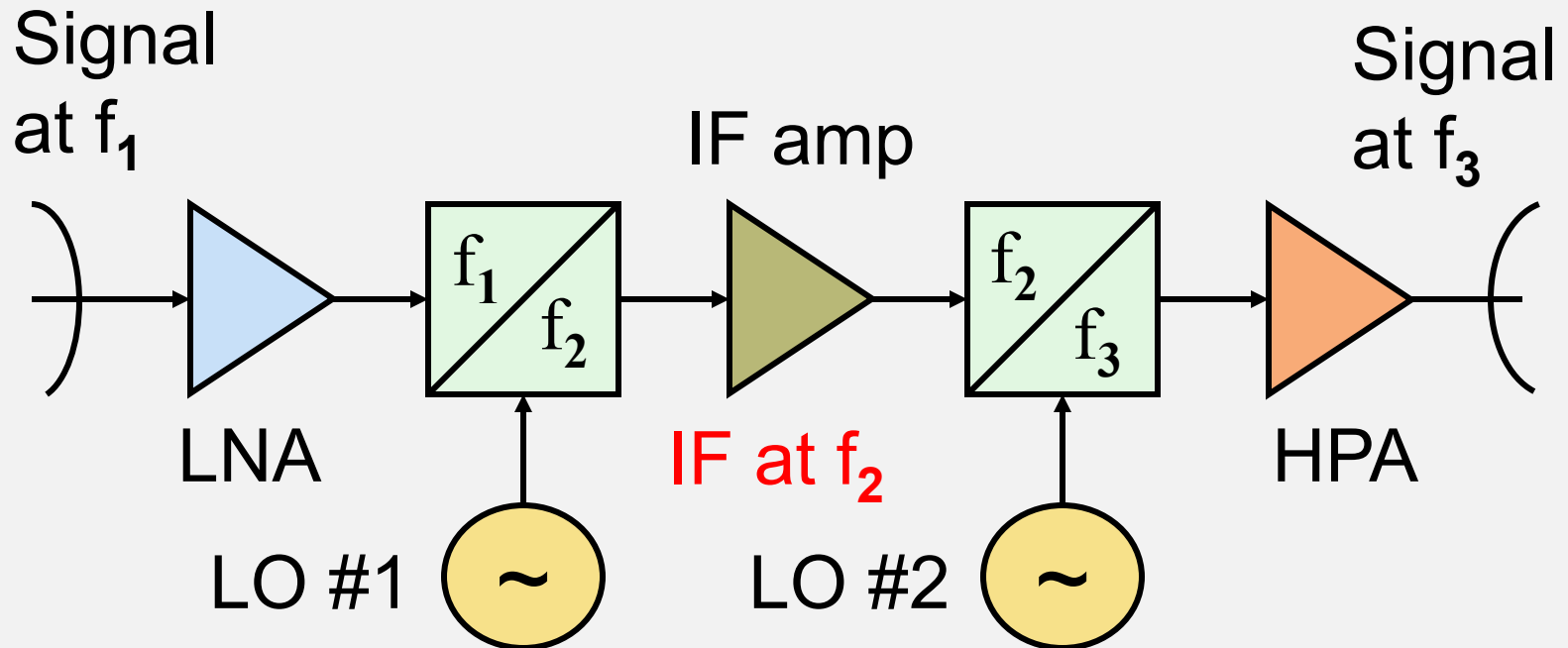
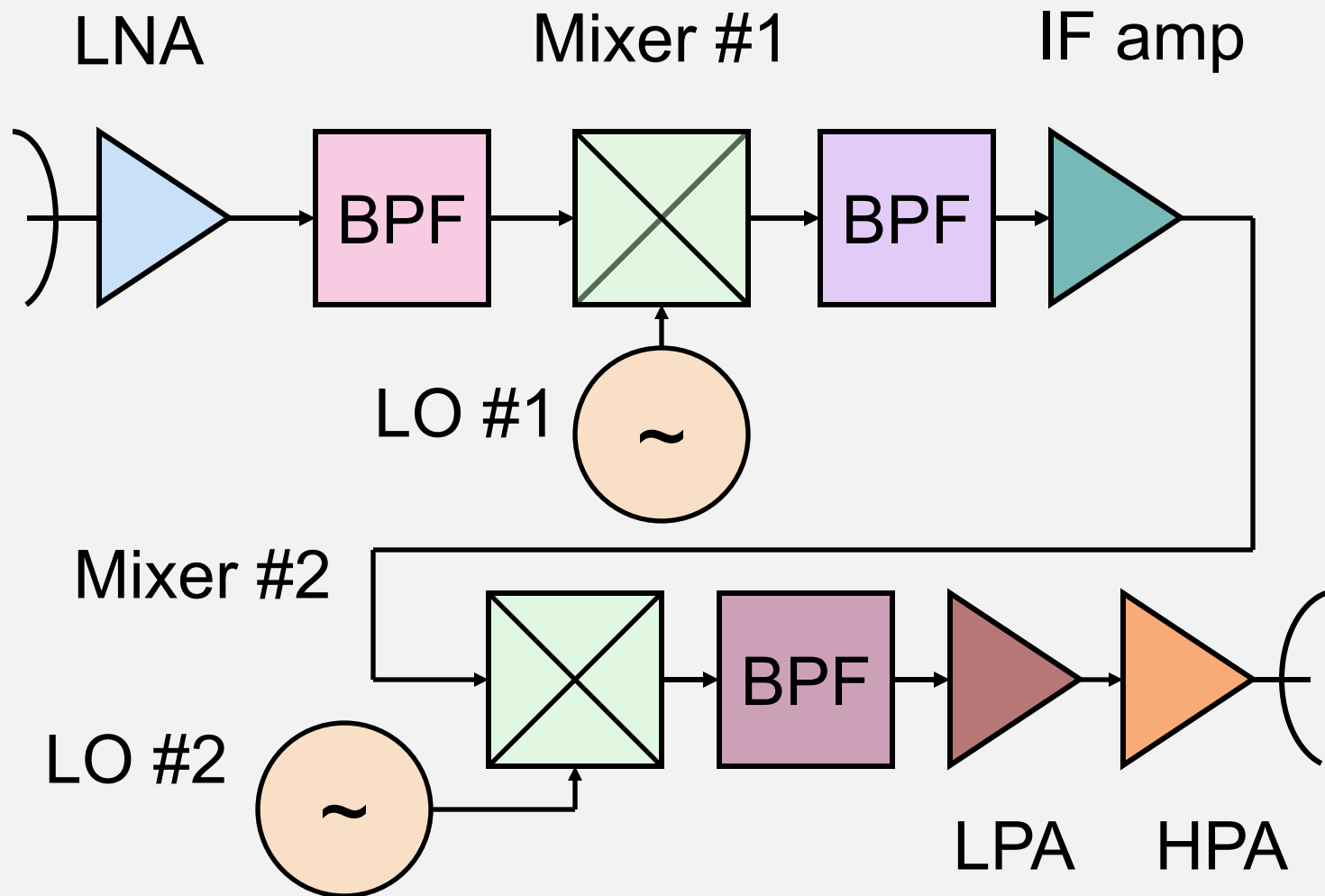


Fig 3.12 - text

Double conversion transponder



Ku-Band Transponder Example

- Uplink frequency: 14.0 GHz
- Downlink frequency: 11.5 GHz
- IF frequency: 1.0 GHz
- What are local oscillator frequencies for low side injection?
- *Low side injection* means LO frequency is below signal frequency

Ku-Band Transponder Example

- LO frequency is difference between signal frequency and IF frequency

- For low side injection

$$f_{LO} = f_{\text{signal}} - f_{IF}$$

- First LO frequency for downconversion is

$$f_{LO1} = 14.0 - 1.0 = 13.0 \text{ GHz}$$

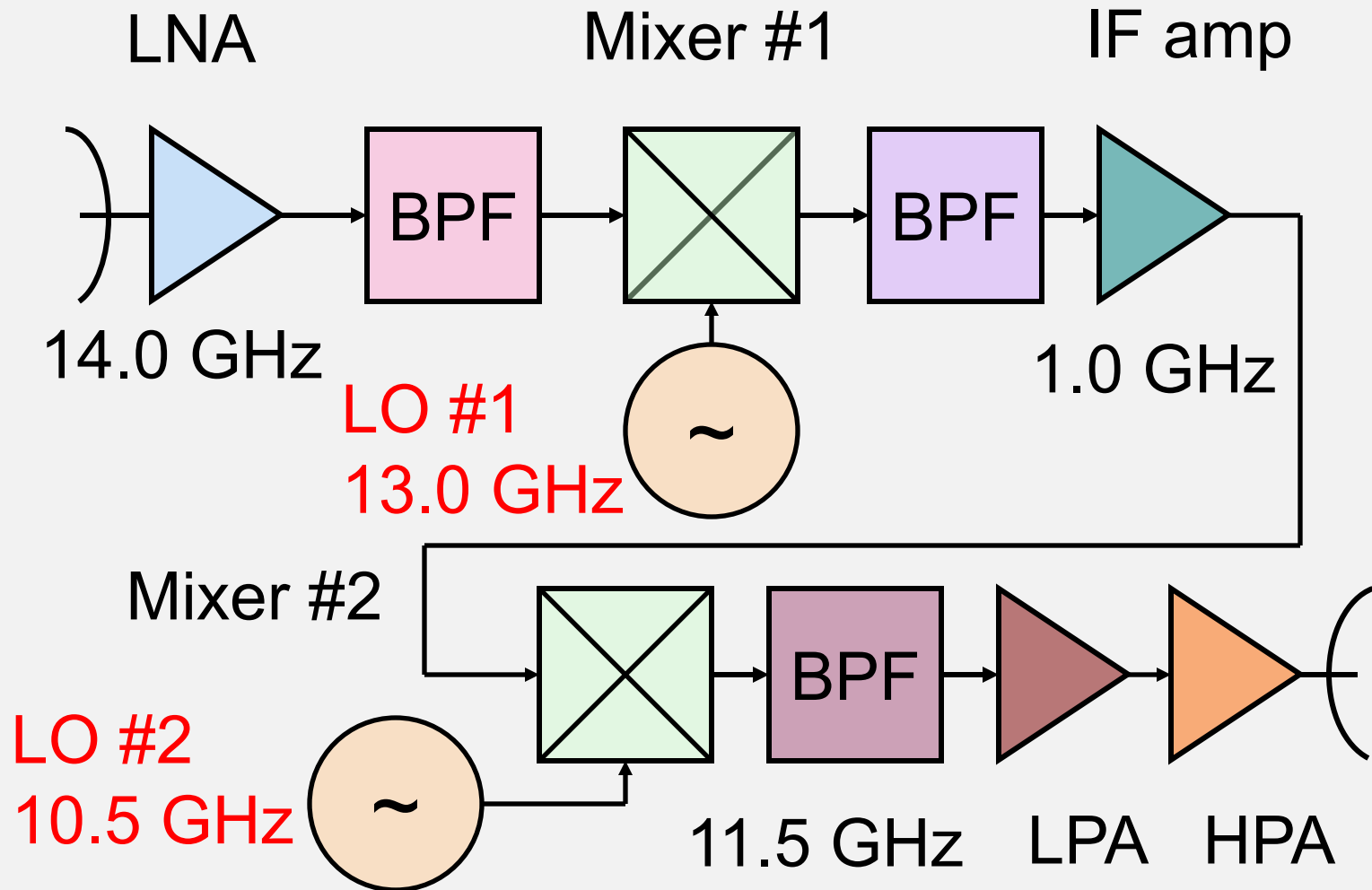
- Second LO frequency for upconversion is

$$f_{LO2} = 11.5 - 1.0 = 10.5 \text{ GHz}$$

Note on Conversion:

- *Downconversion* refers to transformation of signal at the input frequency to signal at IF
- *Upconversion* refers to transformation of signal at IF to signal at the output frequency
- Usually, $f_{\text{in}} > \text{IF} < f_{\text{out}}$
- E.g., 14.0 GHz $\rightarrow$ 1.0 GHz $\rightarrow$ 11.5 GHz

Double conversion transponder - low side injection



Ku-Band Transponder Example

- We could have set the LO frequencies for high side injection (low side is usually preferred). For high side injection

$$f_{LO} = f_{\text{signal}} + f_{IF}$$

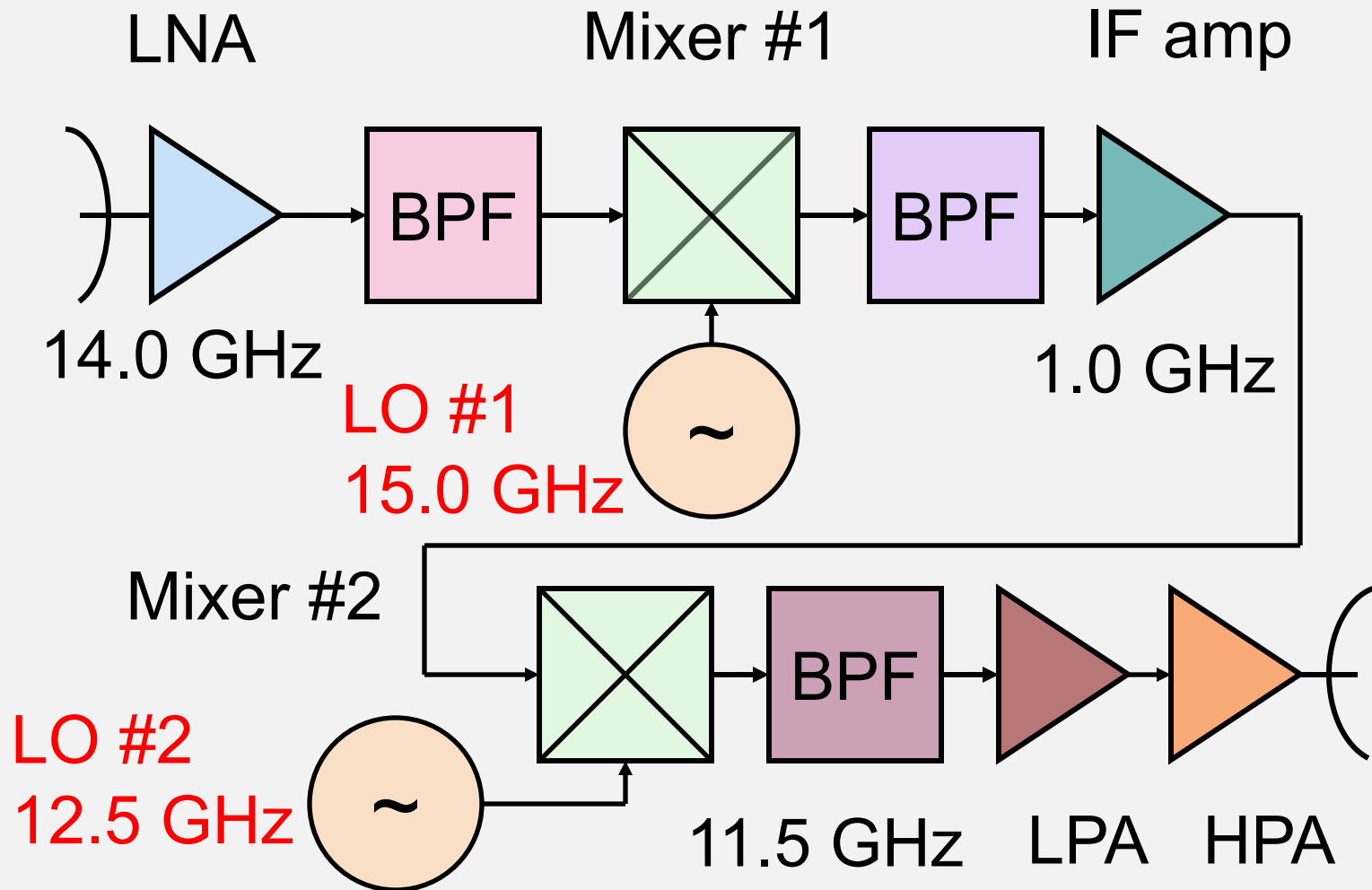
- First LO frequency for *downconversion* is

$$f_{LO1} = 14.0 + 1.0 = 15.0 \text{ GHz}$$

- Second LO frequency for *upconversion* is

$$f_{LO2} = 11.5 + 1.0 = 12.5 \text{ GHz}$$

Double conversion transponder – high side injection



Superheterodyne systems

- The translation of a carrier frequency to another carrier frequency by mixing is called *heterodyning*
- Mixing to an intermediate frequency is called *superheterodyning*
- Signal processing (filtering, amplification) is more easily carried out at the IF frequency
- The principle is applied in virtually all communication systems (see handout)

Image Rejection

- All superhet receivers need image rejection filters
- Image channel is another set of frequencies that can reach the IF amplifier
- Image frequency is $2 \times \text{IF}$ away from signal
- Image signal is below signal frequency for low side injection

Frequency Downconversion ($f_s \rightarrow f_{IF}$)

- With low side injection we set

$$f_{LO} = f_s - f_{IF}$$

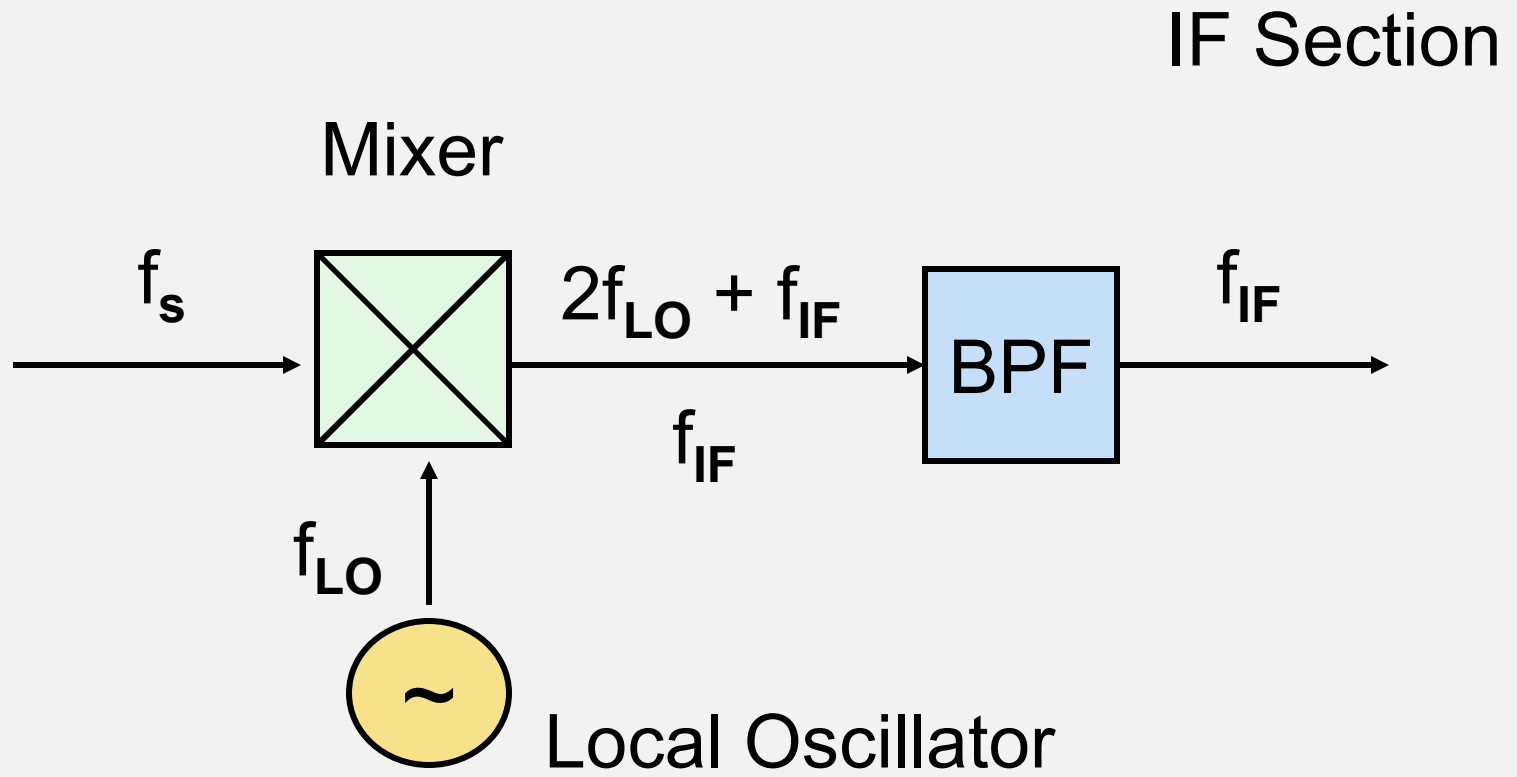
- So $f_s = f_{LO} + f_{IF}$
- Mixing occurs between f_s and f_{LO} :

$$f_s \times f_{LO} = f_s \pm f_{LO}$$

$$\text{(Sum)} \quad f_s + f_{LO} = 2f_{LO} + f_{IF} \quad (\text{remove with filter})$$

$$\text{(Diff)} \quad f_s - f_{LO} = f_{IF} \quad (\text{pass to IF section})$$

Frequency Conversion: Low side Injection



Frequency Conversion: Low side Injection

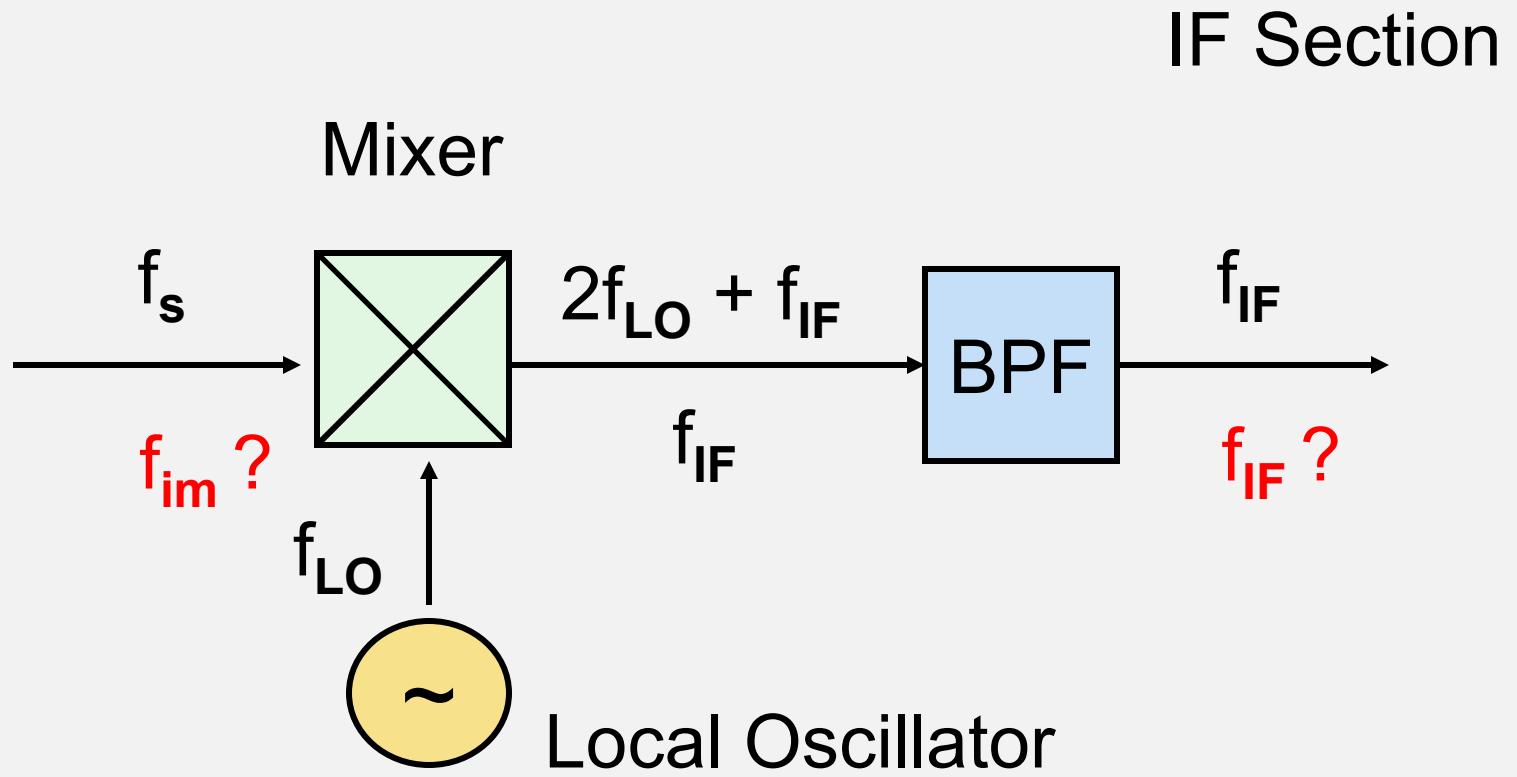


Image frequency

- Is there another frequency, f_{im} , that translates to the same f_{IF} ?
- Try $f_{im} = f_{LO} - f_{IF}$ (Compare: $f_s = f_{LO} + f_{IF}$)
- Mixing occurs between f_{im} and f_{LO} :
$$f_{im} \times f_{LO} = f_{im} \pm f_{LO}$$

(Sum) $f_{im} + f_{LO} = 2f_{LO} - f_{IF}$ (remove with filter)

(Diff) $f_{im} - f_{LO} = -f_{IF}$ (passed to IF section)
- This other frequency, f_{im} , has passed to the IF section, the same as the signal frequency, f_s !

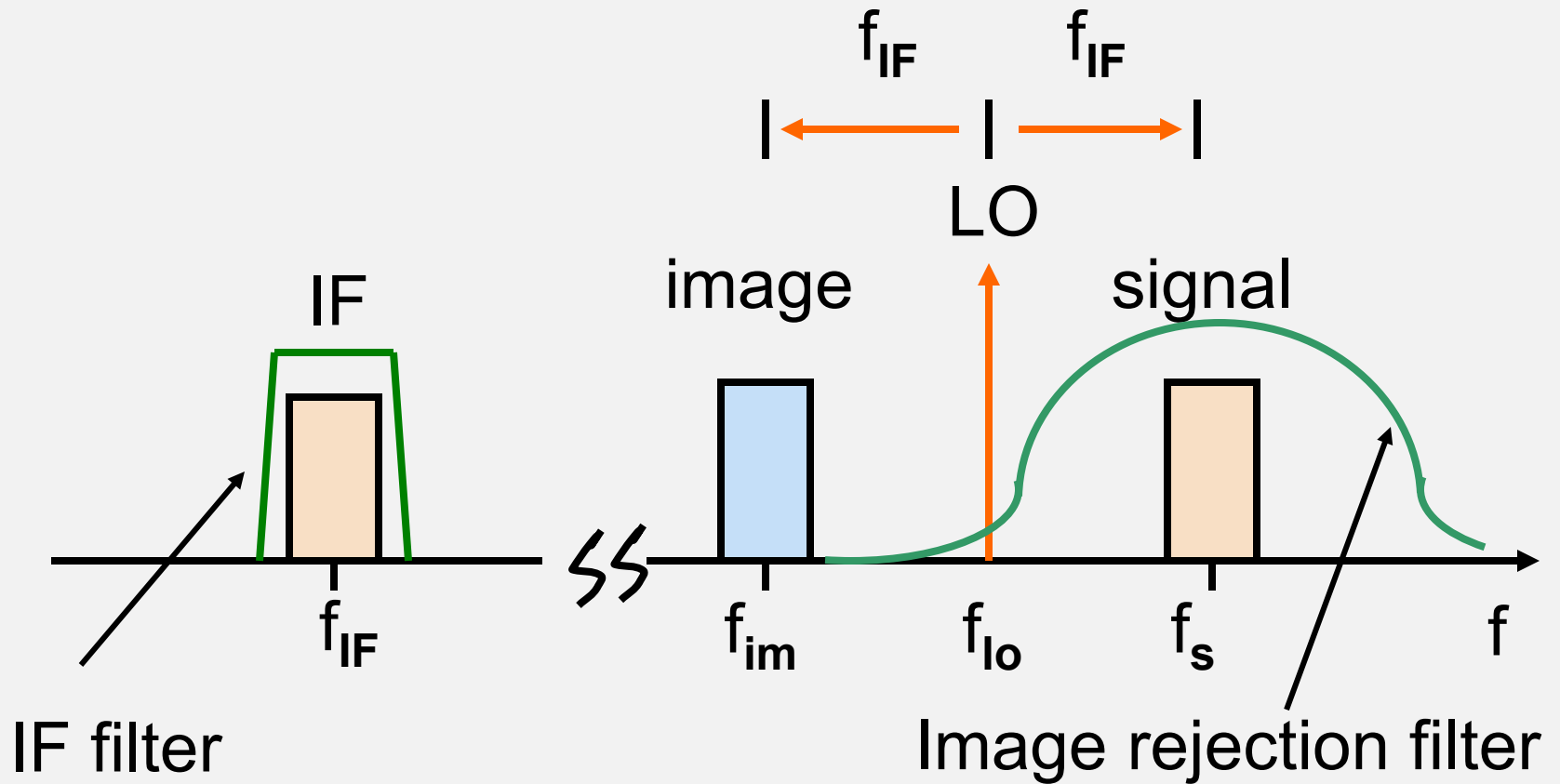
Image frequency

- The frequency, f_{im} , is called the image frequency and it will interfere with processing of the desired signal
- The image frequency needs to be blocked by an *image rejection filter* placed before the mixer
- Why not after the mixer?

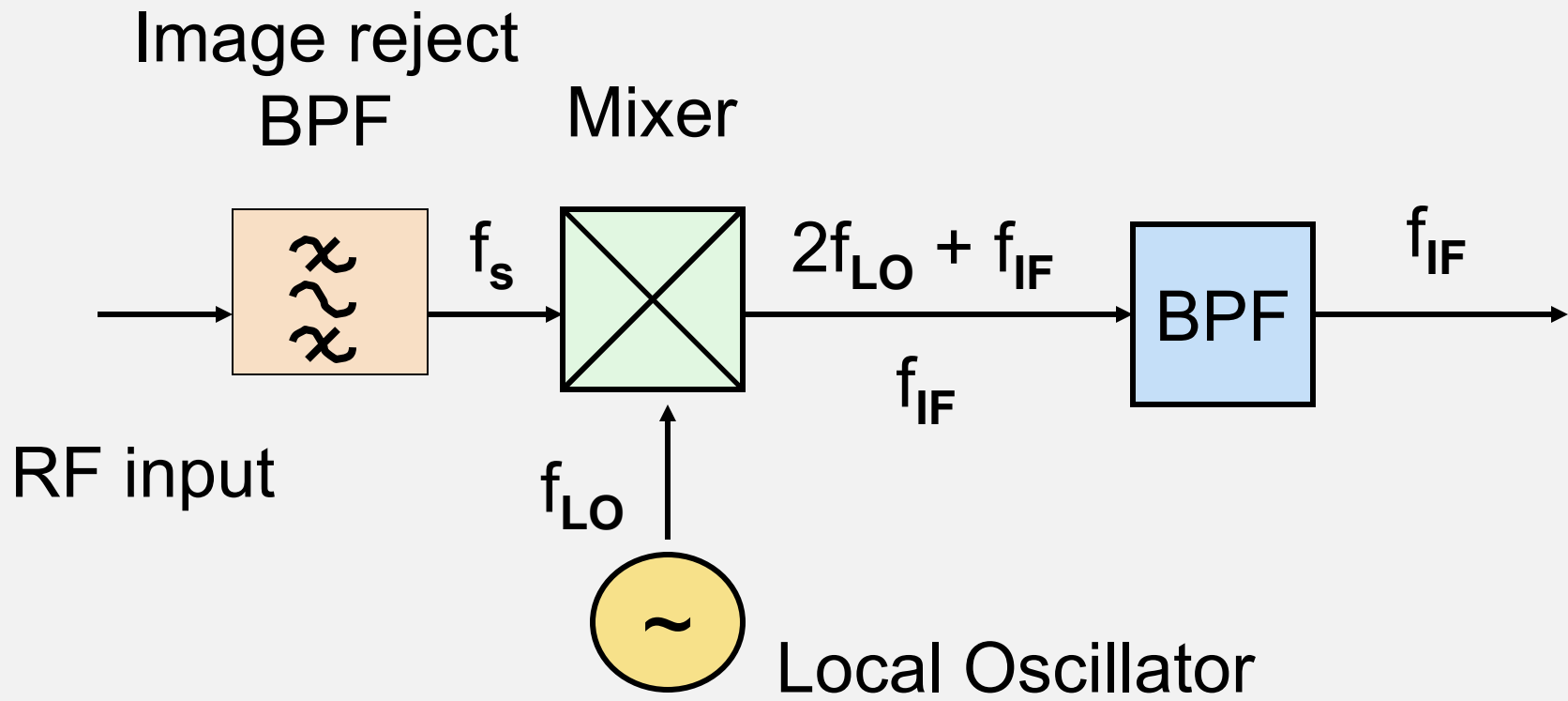
Image frequency

- The frequency, f_{im} , is called the image frequency and it will interfere with processing of the desired signal
- The image frequency needs to be blocked by an *image rejection filter* placed before the mixer
- Why not after the mixer?
- Because that would be too late...both f_s and f_{im} will have mapped to the same f_{IF} and now can't be separated

Receiver spectra



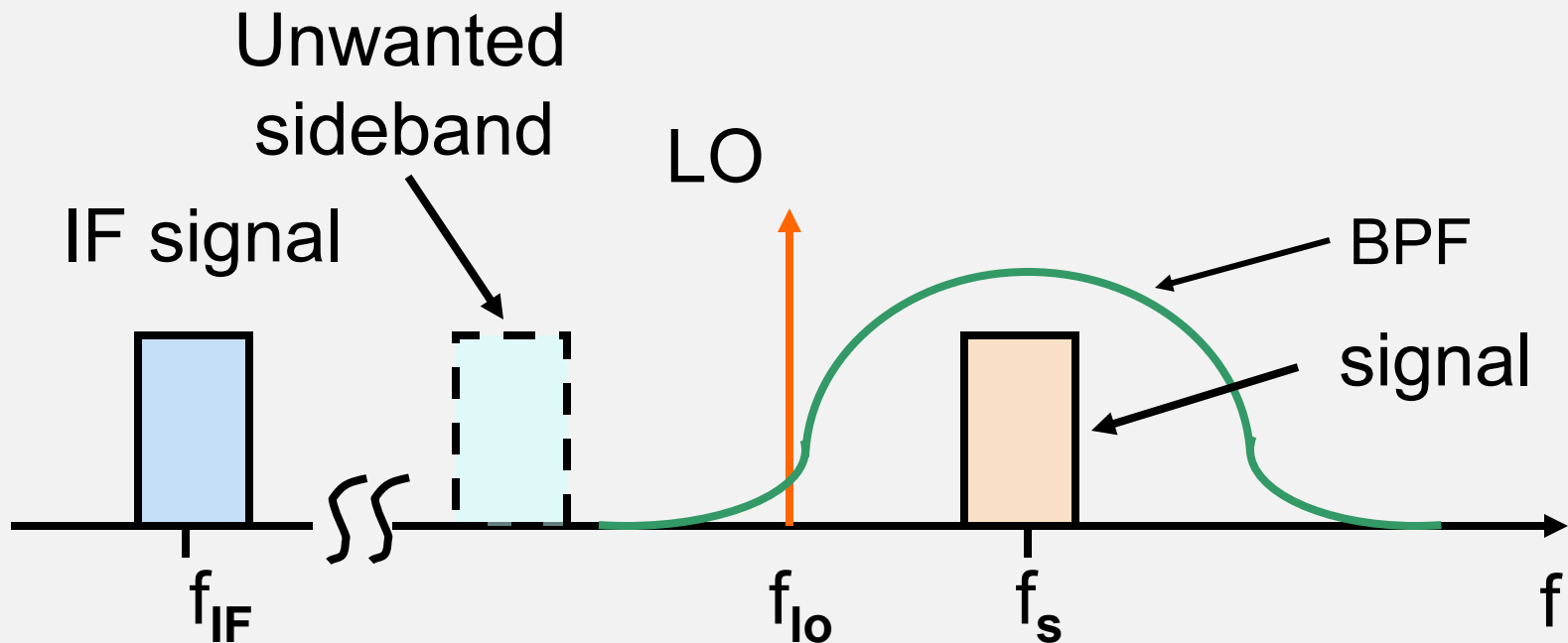
Frequency Downconversion: Low side Injection with Image Rejection

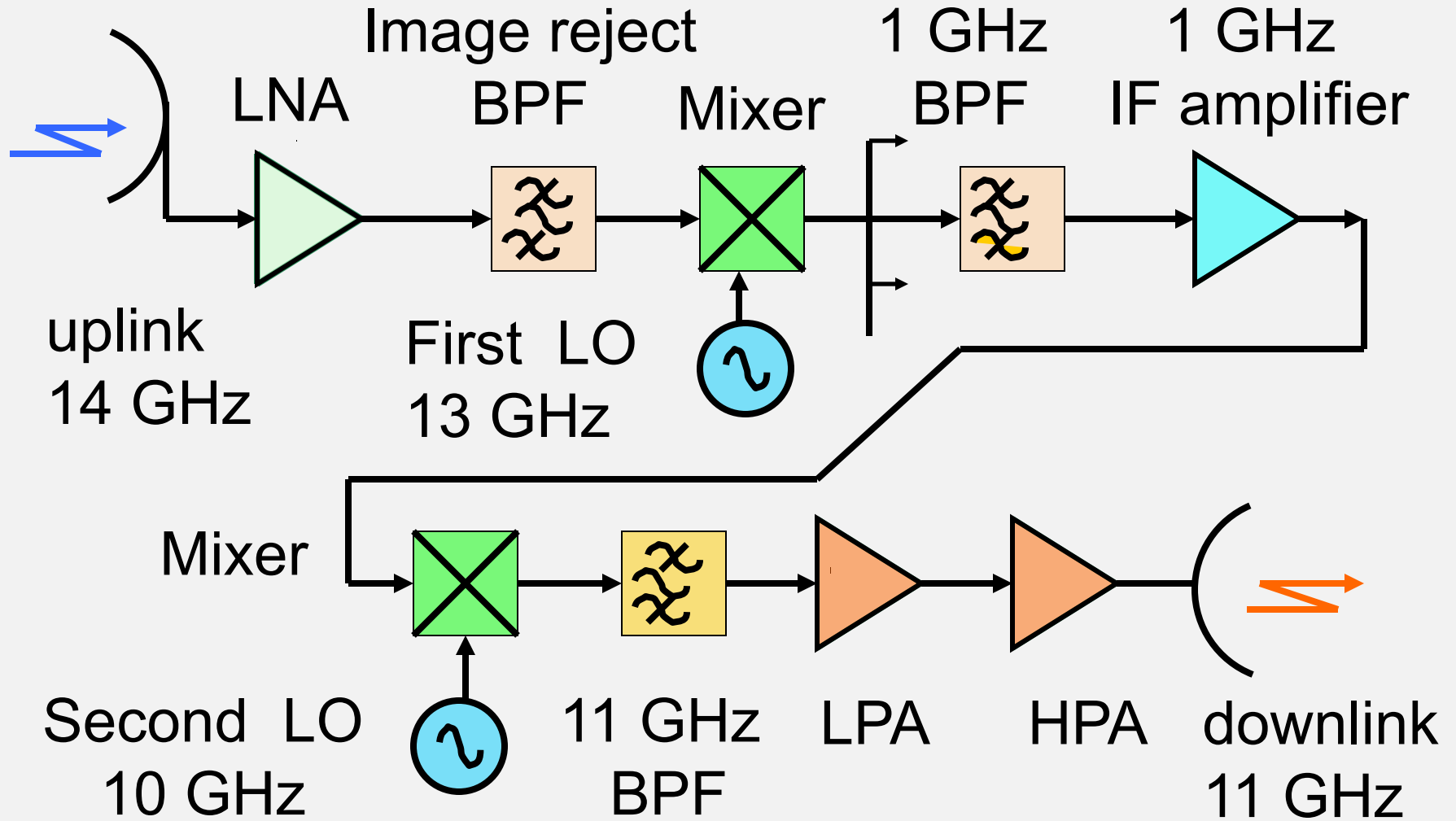


Frequency Upconversion ($f_{IF} \rightarrow f_s$)

$f_{LO} \times f_{IF} = f_{LO} \pm f_{IF}$ leading to

- desired signal at $f_s = f_{LO} + f_{IF}$
- unwanted *sideband* at $f_x = f_{LO} - f_{IF}$

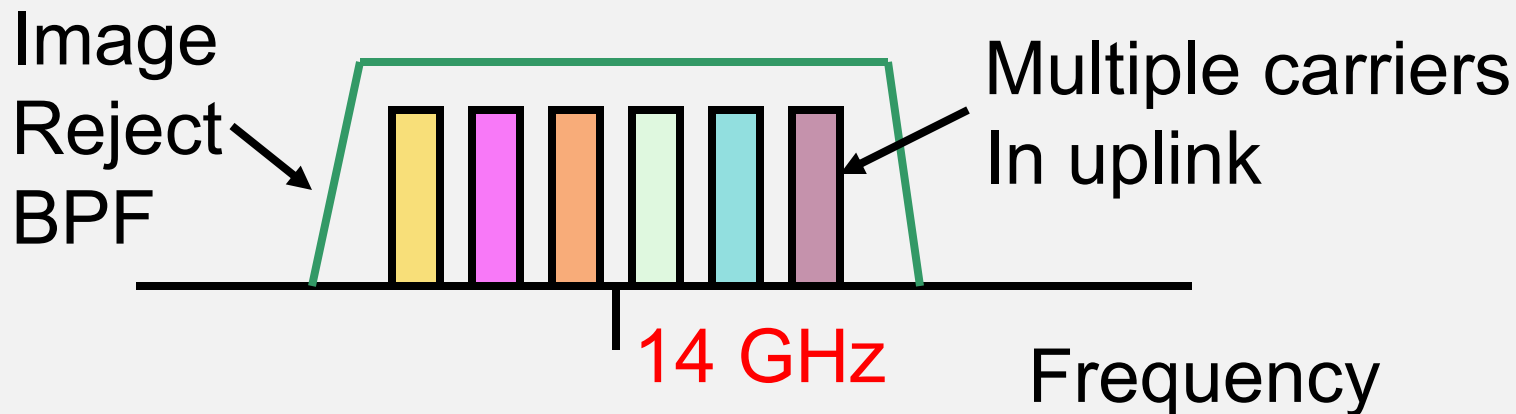




Ku-band Bent Pipe Transponder

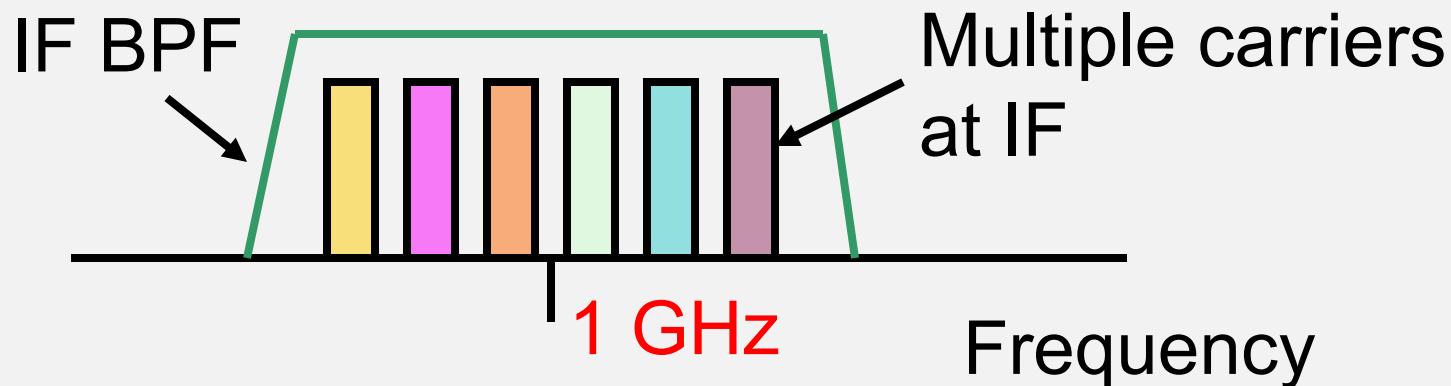
Channels and Transponders

- The uplink and downlink transmissions carry signals within discrete channels
- Consider a 14 GHz uplink frequency and a 500 MHz bandwidth with 36 MHz channels (40 MHz separation) → 12 channels / 12 transponders



IF Amplifier

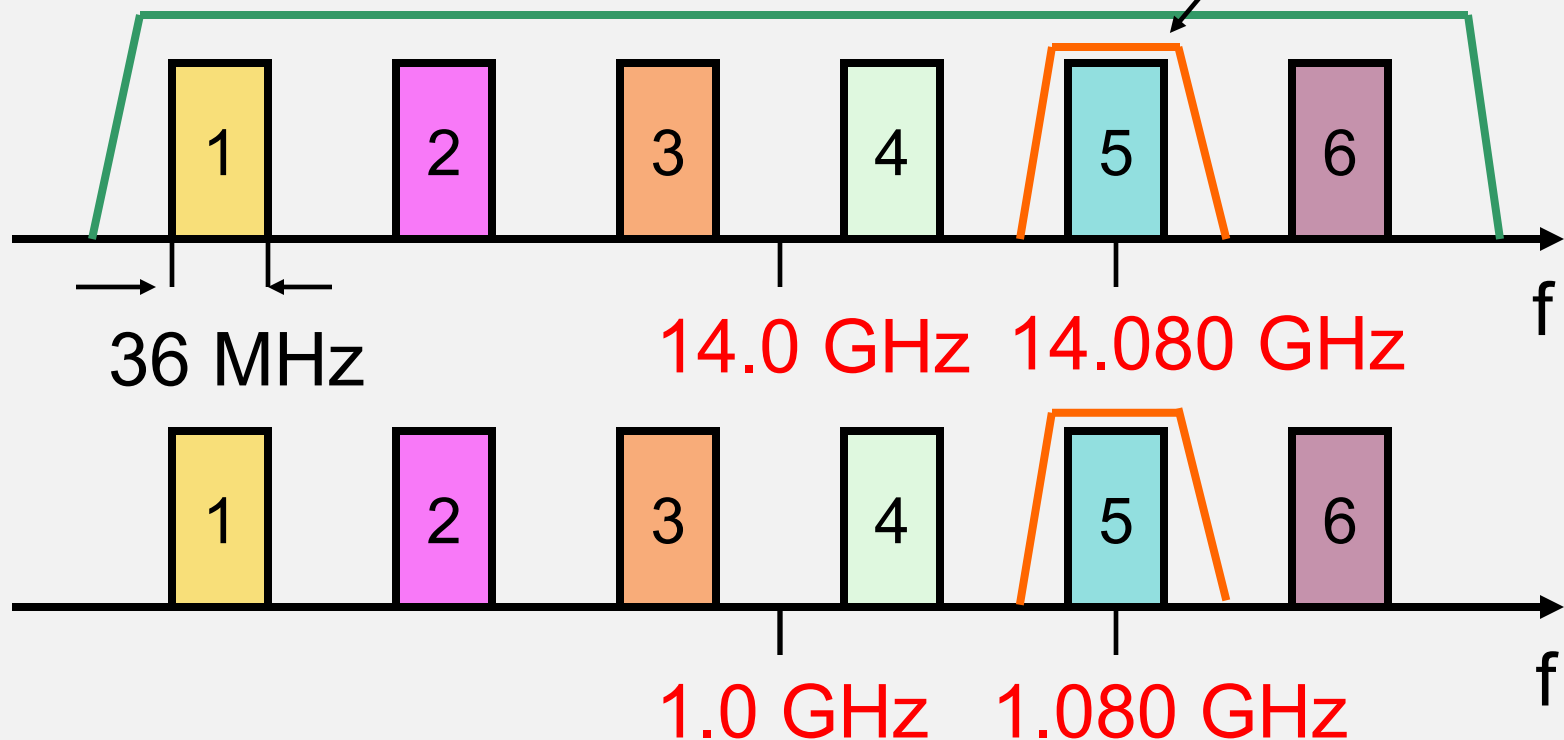
- The channels map 1 for 1 to the IF stage
- Each channel is tapped and processed by one transponder
- After downconversion:



Channel Selection

Multiple carriers
in 14 GHz uplink

Channel select filter

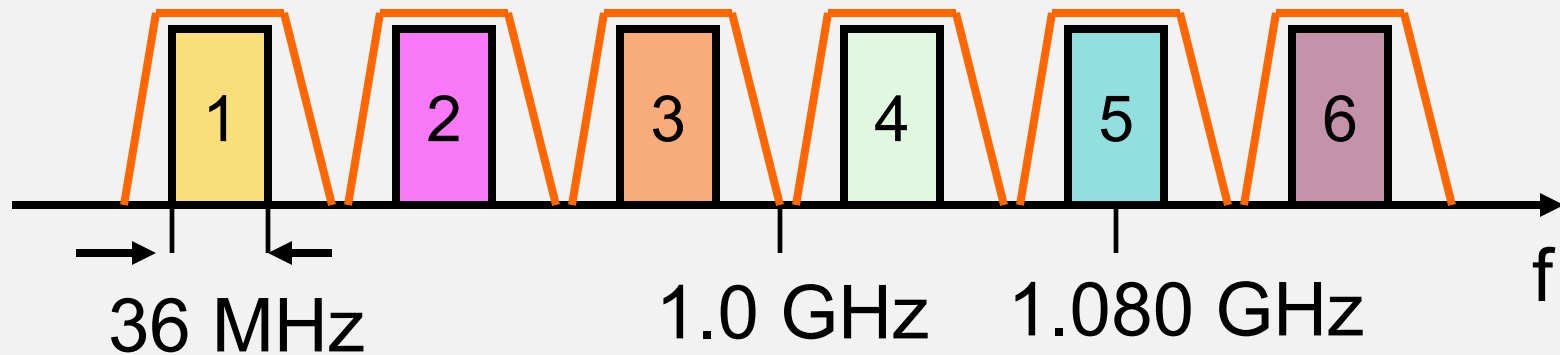


Downconversion of multiple channels in receiver

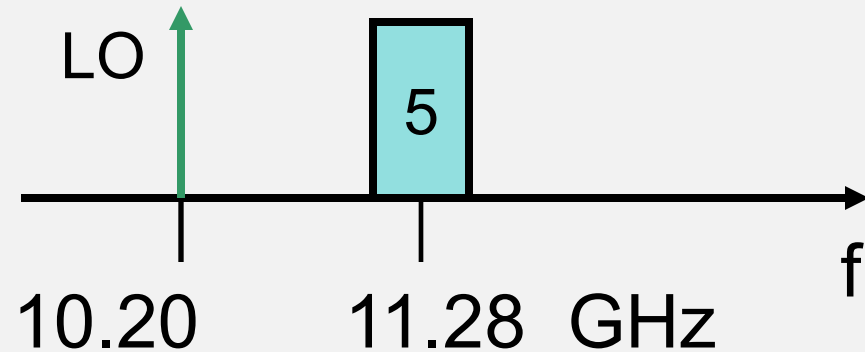
Channel Selection

Multiple carriers
in 1 GHz IF

Channel select
filters



Upconversion
of single channel
in one transponder

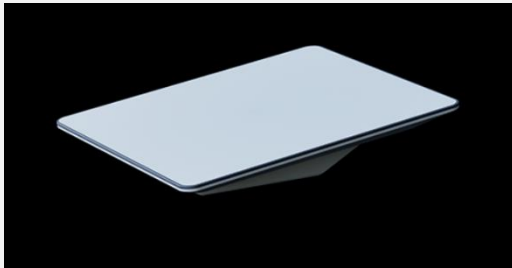


Antennas

- Often the critical element in a radio communication system
- Gain and beamwidth are related
 - High gain $\Rightarrow$ Narrow beam
 - Wide beam $\Rightarrow$ Low gain
- Control and command systems - wide beams
- Communications links - narrow beams

Types of Antennas

- Resonant (wire antennas, < 1 GHz)
- Aperture (horn antennas, > 1 GHz)
- Reflectors (parabolic dishes, > 5 GHz)
- Phased array (> 10 GHz)



Antennas

- TTC&M antennas need to be omni-directional
 - Monopoles and dipoles (wires)
- Communications antennas are directional
 - Horns and reflector antennas
 - Horns: 10 – 25 dB gain
 - Reflectors (dishes) 20 – 60 dB gain
 - Phased arrays

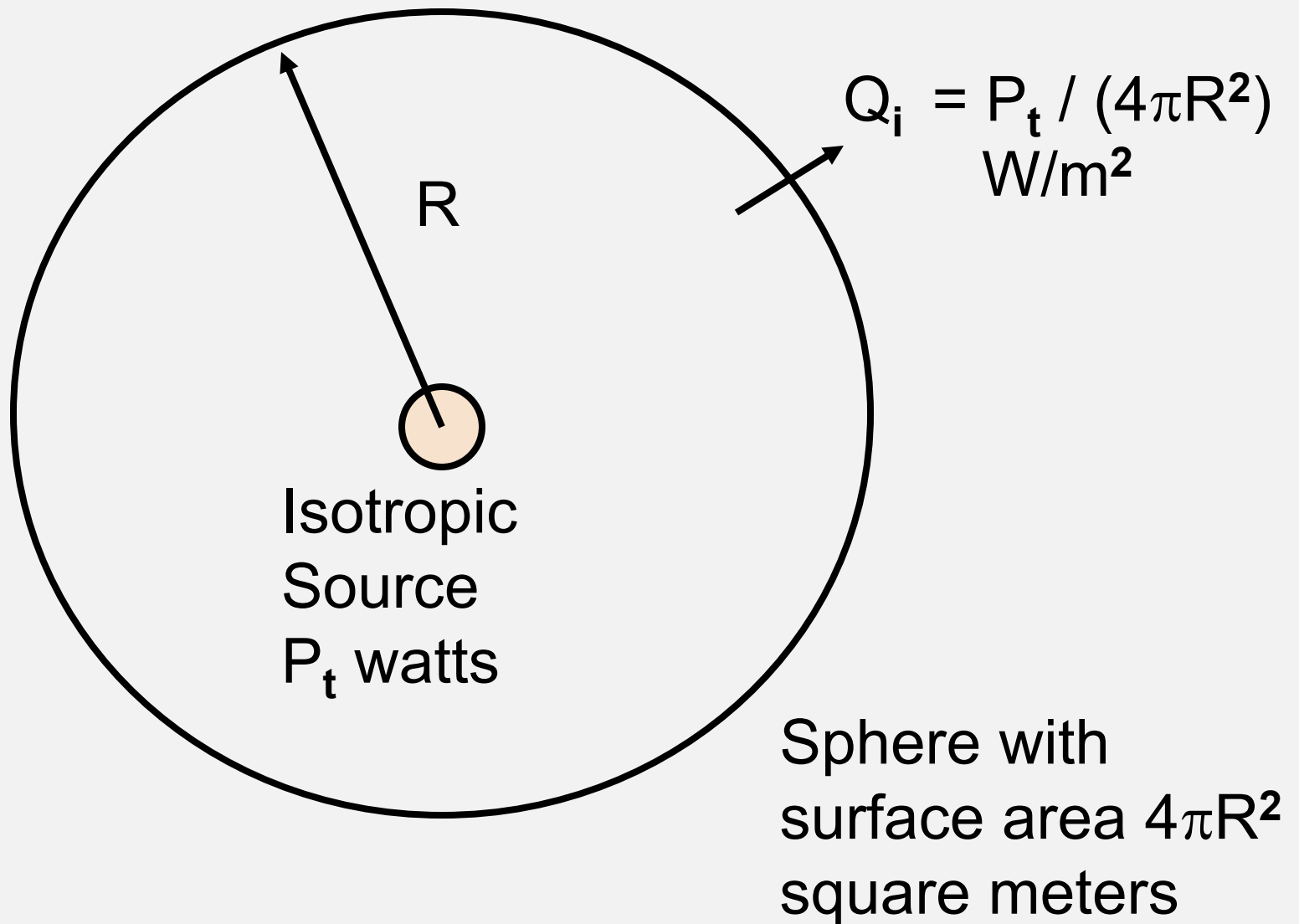
Antennas

Definition of antenna gain:

The increase in received power at a given point with the test antenna relative to the power received from an isotropic antenna

Definition of an isotropic antenna:

An antenna that radiates equally in all directions - does not exist



Power Density from an Antenna

- An isotropic source with power P_t watts would create a power density of Q_i watts / m^2 at a distance R (Eqn. 2.1)

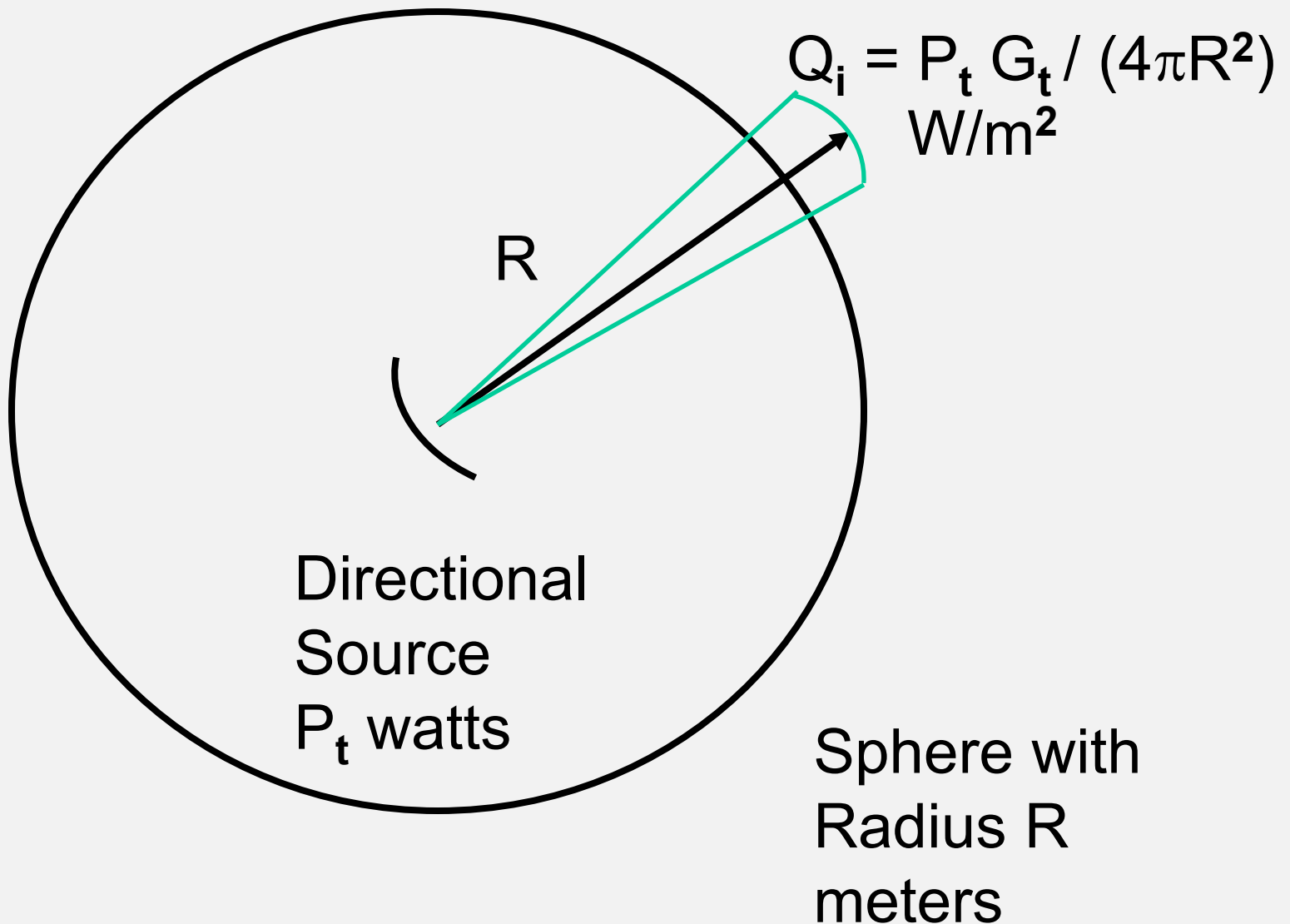
$$Q_i = P_t / (4 \pi R^2) \quad W/m^2$$

Power Density from an Antenna

- If we replace the isotropic antenna with a directive antenna with *gain* G_t in a given direction then the power density in that direction is

$$Q_i = P_t G_t / (4 \pi R^2) \quad \text{W/m}^2$$

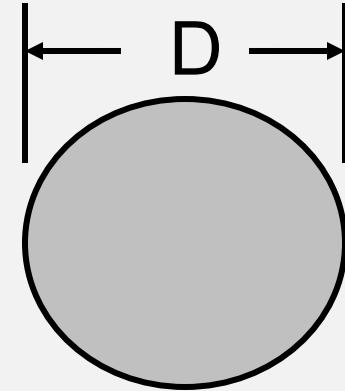
- Note that *gain* generally refers to the gain in the direction of peak antenna performance



Antenna Formulas

- Gain [Eqn 3.1]

$$G = \eta_A \frac{4\pi A}{\lambda^2}$$



$$A = \pi D^2 / 4$$

where η_A is the aperture efficiency of the antenna

A is the physical aperture area
and λ is the wavelength

- $G = \eta_A \times (\pi D / \lambda)^2$ for a circular aperture
- Equations give G as a ratio – not in dB

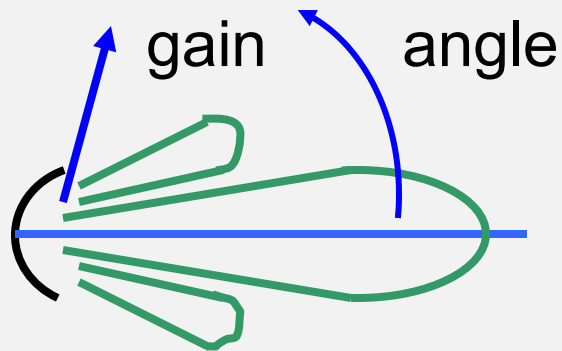
Antennas

- The beamwidth, $\theta_{3\text{dB}}$, is defined as the angular extent between the 3 dB points of the main beam
- Beamwidth can be estimated from [Eqn 3.3] or [Eqn 3.4], e.g.,

$$\theta_{3\text{dB}} \approx 75 \lambda / D \quad [\text{deg}] \quad (\text{Eqn 3.3})$$

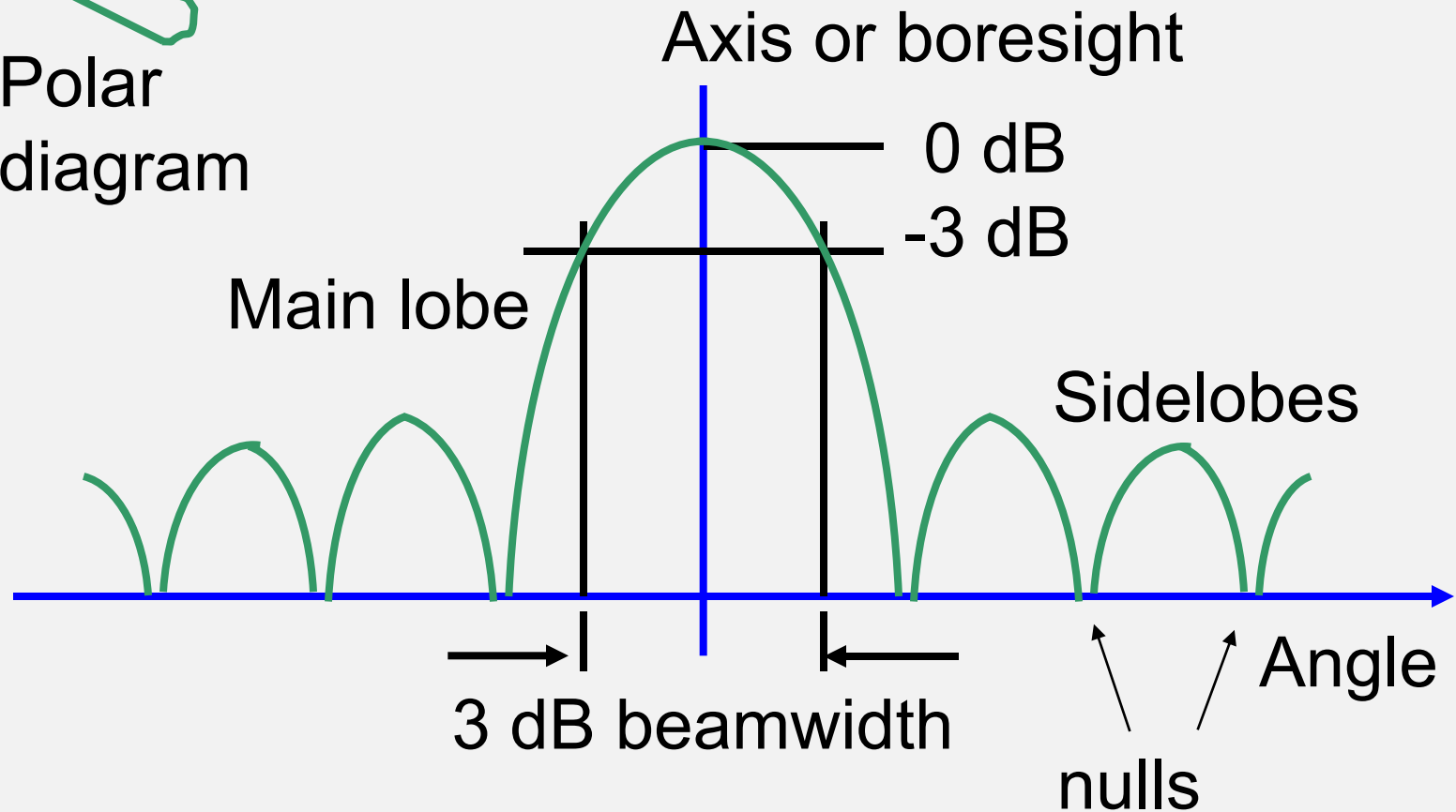
where D = aperture diameter

- If the aperture has two diameters, D_H and D_V , then there are two beamwidths, θ_H and θ_V



Polar
diagram

Antenna Patterns



Antennas

- Antenna gain varies with angle
- If quoted as just “gain”, the maximum (on axis, boresight) value is inferred.
- The variation of antenna gain with angle, $G(\theta)$, is known as the *antenna pattern*
- Antenna patterns have a *main lobe* and *sidelobes*
- Usually presented as gain vs angle in a plane

Antennas

- Antenna approximate relationships

$$G \approx 33,000 / (\theta_1 \times \theta_2)$$

θ_1 , θ_2 are 3 dB beamwidths (in degrees)
for rectangular or elliptical aperture

- $G \approx 33,000 / (\theta_{3 \text{ dB}})^2$

for a circular aperture (Eqn 3.4)

G is a ratio – not in dB

Antenna Example 1

- These approximate relationships are very useful for antenna analysis
- Example: What are the beamwidth and gain of a circular antenna that has $D = 5$ m and $f = 3$ GHz?

Antenna Example 1

- These approximate relationships are very useful for antenna analysis
- Example: What are the beamwidth and gain of a circular antenna that has $D = 5$ m and $f = 3$ GHz?

$$f = 3 \text{ GHz} \rightarrow \lambda = 10 \text{ cm} = 0.1 \text{ m (S band)}$$

$$\theta_{3 \text{ dB}} \approx 75 \lambda / D = 75 (.1 / 5) = 1.5 \text{ deg}$$

$$G \approx 33,000 / (\theta_{3 \text{ dB}})^2 = 14700 \sim 41 \text{ dB}$$

Frequency – Wavelength Conversion

- Frequency and wavelength are related through $c = f \times \lambda$
- It is easier to convert using a form of the equation suitable for satellite communications

$$f \text{ [GHz]} \times \lambda \text{ [cm]} = 30$$

- i.e., the ‘magic number’ is 30 with frequency in GHz and wavelength in centimeters (the usual units)
- Advisable to commit to memory

Antenna Example 2

- A circular VSAT antenna must have a gain of 40 dB at a frequency of 11.5 GHz (Ku band)
 - what is the antenna beamwidth, θ ?
 - what is the antenna diameter, D ?
- Answer:

Use approximate relationship

$$G \approx 33,000 / (\theta_1 \times \theta_2) \text{ with } \theta_1 = \theta_2$$

Antenna Example 2

- 3 dB Beamwidth for circular aperture:

$$G = 40 \text{ dB} = 10,000 \approx 33,000 / (\theta)^2$$

$$\theta = \sqrt{(33,000 / 10,000)} = \sqrt{3.3}$$

$$\underline{\theta = 1.82^\circ}$$

- Antenna diameter (circular aperture)

$$\theta_{3 \text{ dB}} \approx 75 \lambda / D$$

$$D = 75 \lambda / \theta = 75 \times \lambda / 1.82^\circ$$

Antenna Example 2

- Wavelength conversion:

$$f \text{ [GHz]} \times \lambda \text{ [cm]} = 30$$

$$11.5 \text{ [GHz]} \times \lambda \text{ [cm]} = 30$$

$$\begin{aligned} \rightarrow \lambda &= 2.61 \text{ cm} \\ &= 0.0261 \text{ m} \end{aligned}$$

- Antenna diameter (Circular aperture)

$$D = 75 \lambda / \theta = 75 \times \lambda / 1.82 \text{ m}$$

$$\underline{D = 1.08 \text{ m}}$$

Antenna Examples 1 & 2

- Note that these two examples produce antennas with about the same gain, ~ 40 dB
- But the antenna in S-band is 5 m in diameter while the antenna in Ku band is ~ 1 m in diameter
- We can manage with smaller antennas by increasing frequency
- The 'size' of an antenna in terms of antenna performance goes as $1 / \lambda$

Antenna Example 3

If we know (or estimate) antenna efficiency we can find aperture diameter from the exact expression

$$G = \eta_A \times 4 \pi A / \lambda^2$$

For a circular aperture

$$A = \pi D^2 / 4$$

and

$$G = \eta_A \times (\pi D / \lambda)^2$$

Antenna Example 3

So, reworking the previous example

$$D = \sqrt{(G \lambda^2 / \eta_A \pi)}$$

If we have $\eta_A = 0.6$ (60%) then

$$D = \sqrt{(10,000 \times (0.0261)^2 / 0.6 \pi^2)}$$

$$\underline{D = 1.29 \text{ m}}$$

- Note that the two methods give different answers, 1.29 vs. 1.08 m (depends on value of η_A assumed in approximate formula)

Antennas

- Broad beam / low gain antennas
- Mobile telephones use broad beam antennas that do not need to be aimed
- E.g. Satellite telephone

Beamwidth: 360 degrees in azimuth

90 degrees in elevation

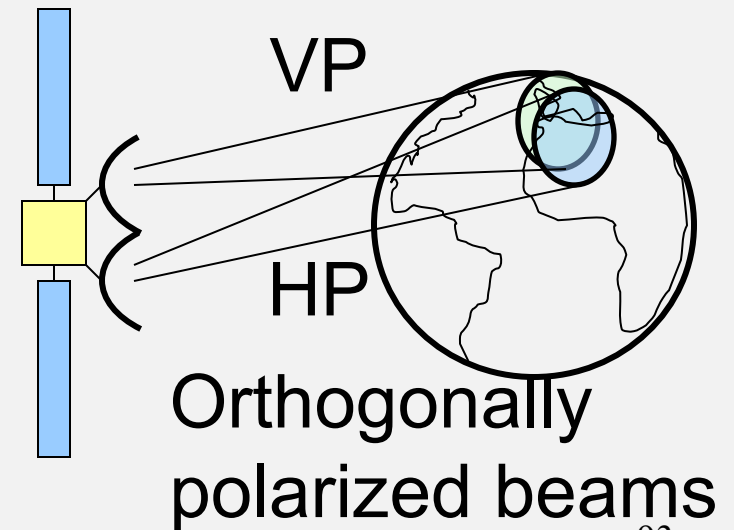
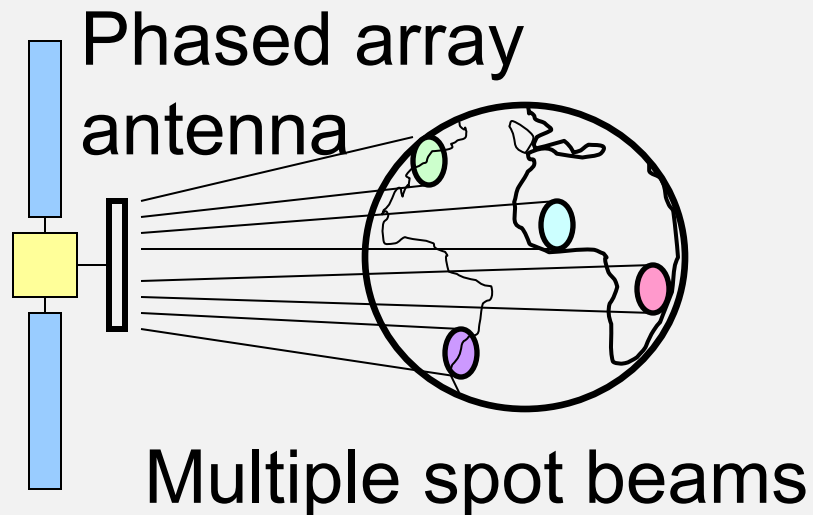
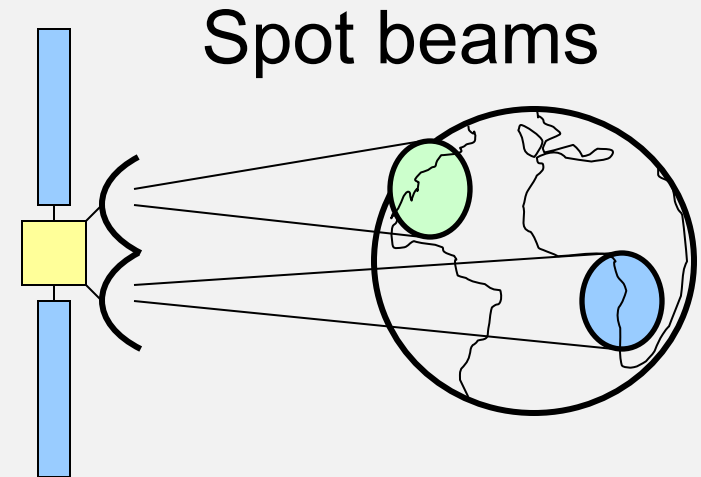
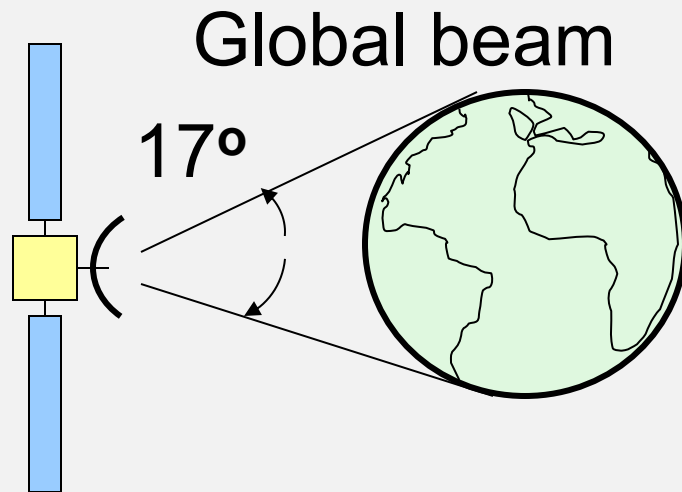
Antennas

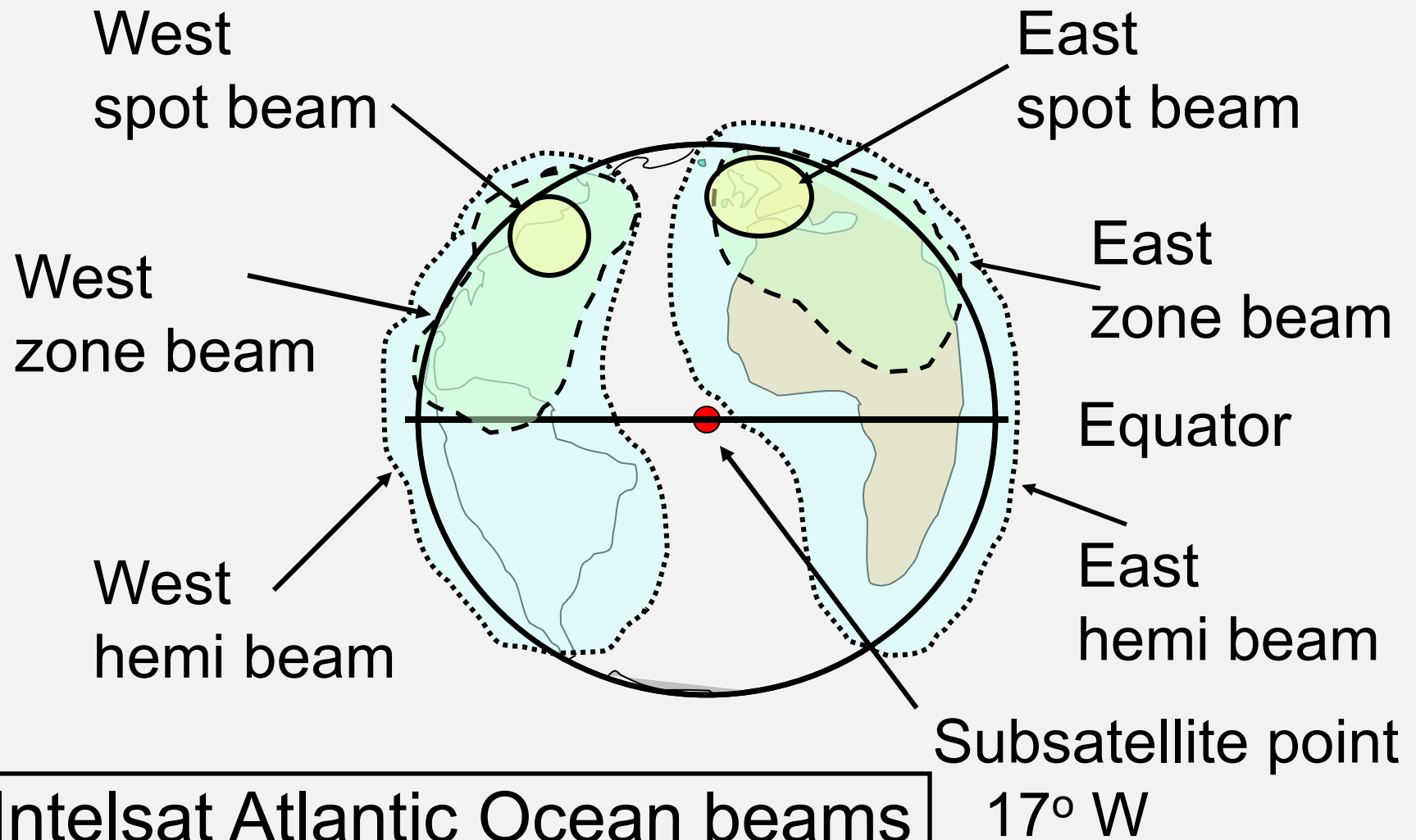
- $G = 33,000 / (360 \times 90) = 1.02 \Rightarrow 0 \text{ dB}$
- Zero dB gain (omni-directional) antennas are problematic in satellite links because they require high transmitter powers
- There are safety limits on transmit power with mobile telephones

Satellite Antennas

- Simplest: Global beam, 17° beam from GEO
- Has low gain - 20 dB on axis, 17 dB at edge
- Regional beams cover continents
- Spot beams cover specific areas
- Frequency reuse options:
 - > Spatial – narrow beams at same frequency
 - > Polarization – overlaid beams with orthogonal polarizations

Fig 3.15 - text





SES-14 Satellite (47.5° W)

- Replaced an older satellite operating in C-band
- Includes a C-band payload and a Ku-band payload
- Ku-band payload includes spot beams for covering the Americas and the North Atlantic region
- Ku-band wide beams cover Latin America, the Caribbean, and across the North Atlantic

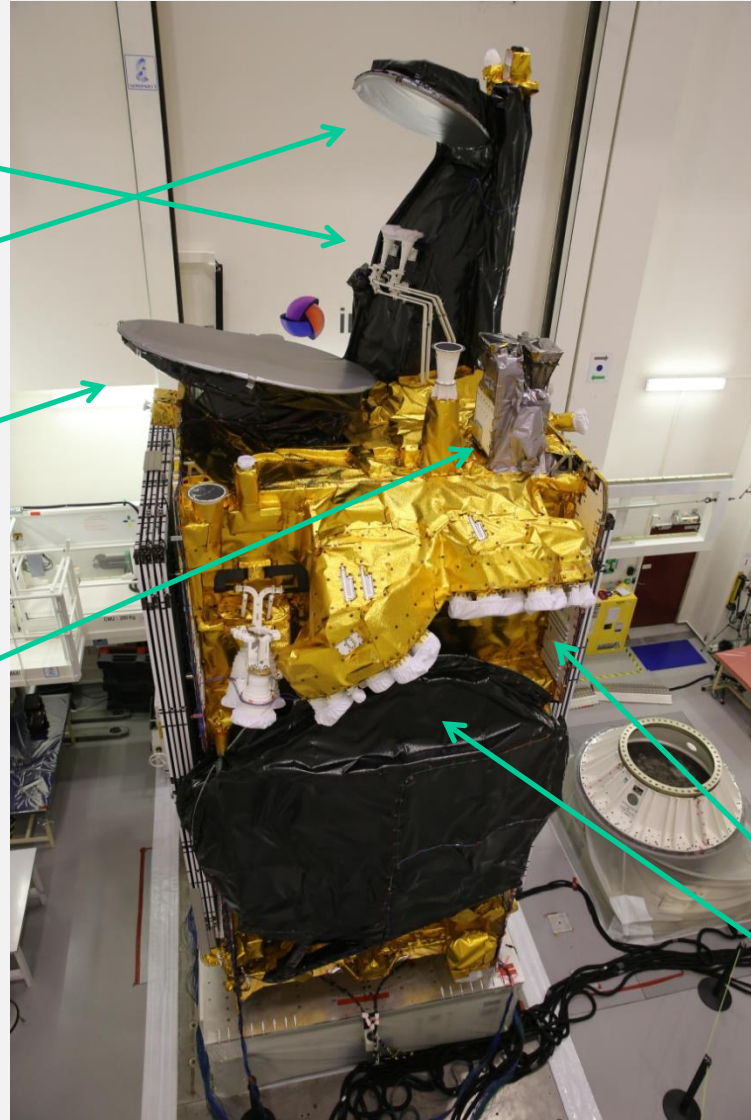
SES-14 Satellite

Feed horns

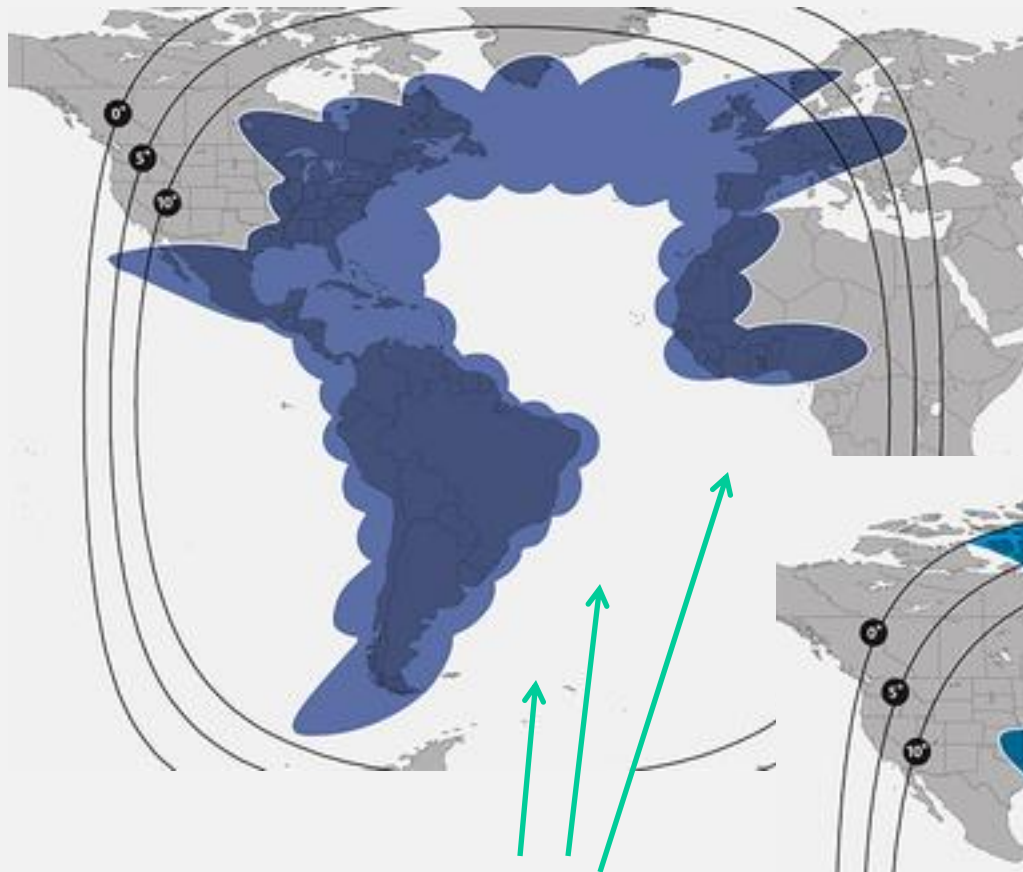
Small dish antenna

Large dish antenna

Includes a NASA
scientific instrument
package: Global-
scale Observations
of the Limb and
Disk (GOLD)



???



SES-14 Ku-band coverage

Wide beams ↓



Spot beams
(requires multiple
horn feeds)

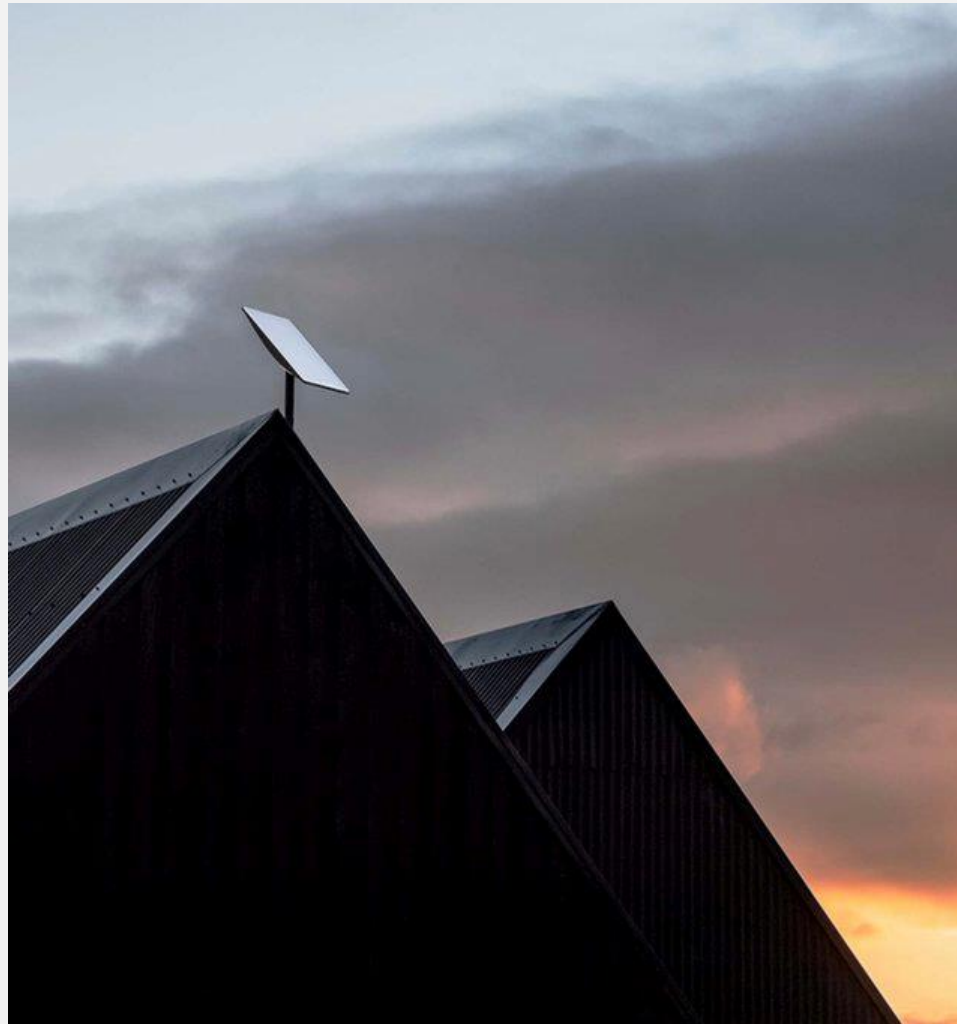
Earth Station Antennas

- Hub antennas are large, fully steerable, and very expensive - \$ 5M
 - > high capacity, multi-channel, high revenue applications
- Consumer antennas need to be very small and low cost – a few hundred dollars
 - > Direct Broadcast Satellite TV
- Starlink basic consumer phased-array antenna costs \$499 to buy and \$1000 to manufacture



Fig 11.4

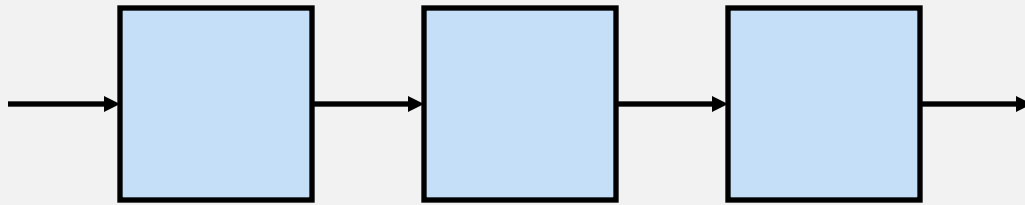
Starlink rectangular phased-array antenna



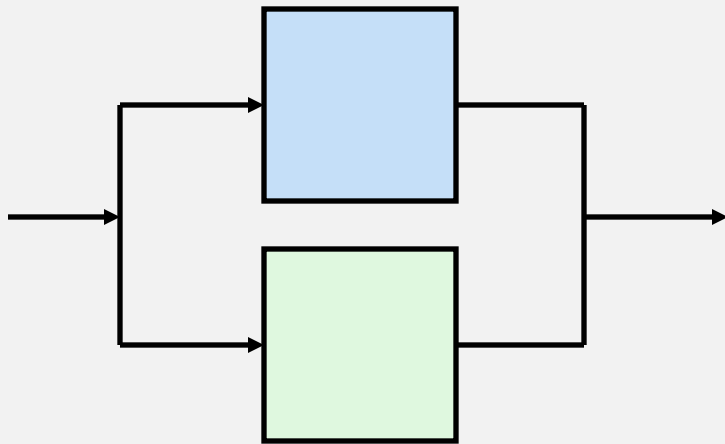
Reliability

- Satellites cannot be easily repaired in orbit
- Reliability and lifetime are important issues
- Some components have limited lifetime
- E.g. High power amplifiers (HPAs)
- *Redundancy* is used to overcome failures
- E.g. HPA may have two amplifiers in parallel
- If one fails the other continues to operate

Redundancy configurations

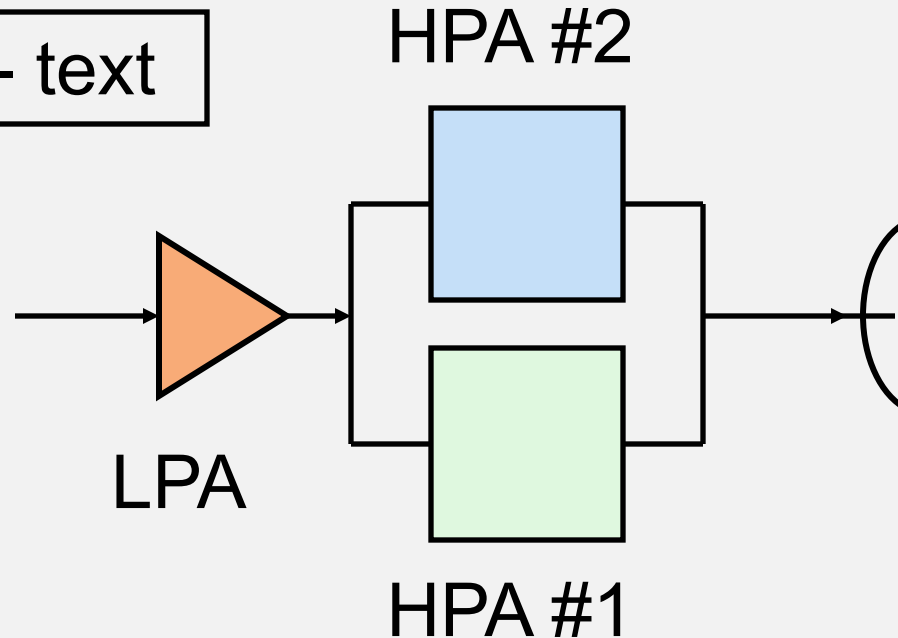


Series redundancy – blocks must fail short circuit



Parallel redundancy – blocks must fail open circuit

Part Fig 3.18 - text



One HPA #2 is off initially, HPA #1 is on.
When #1 fails, #2 is turned on.

Redundant HPAs in output stage of transponder

Components are subjected to a burn-in period before installation

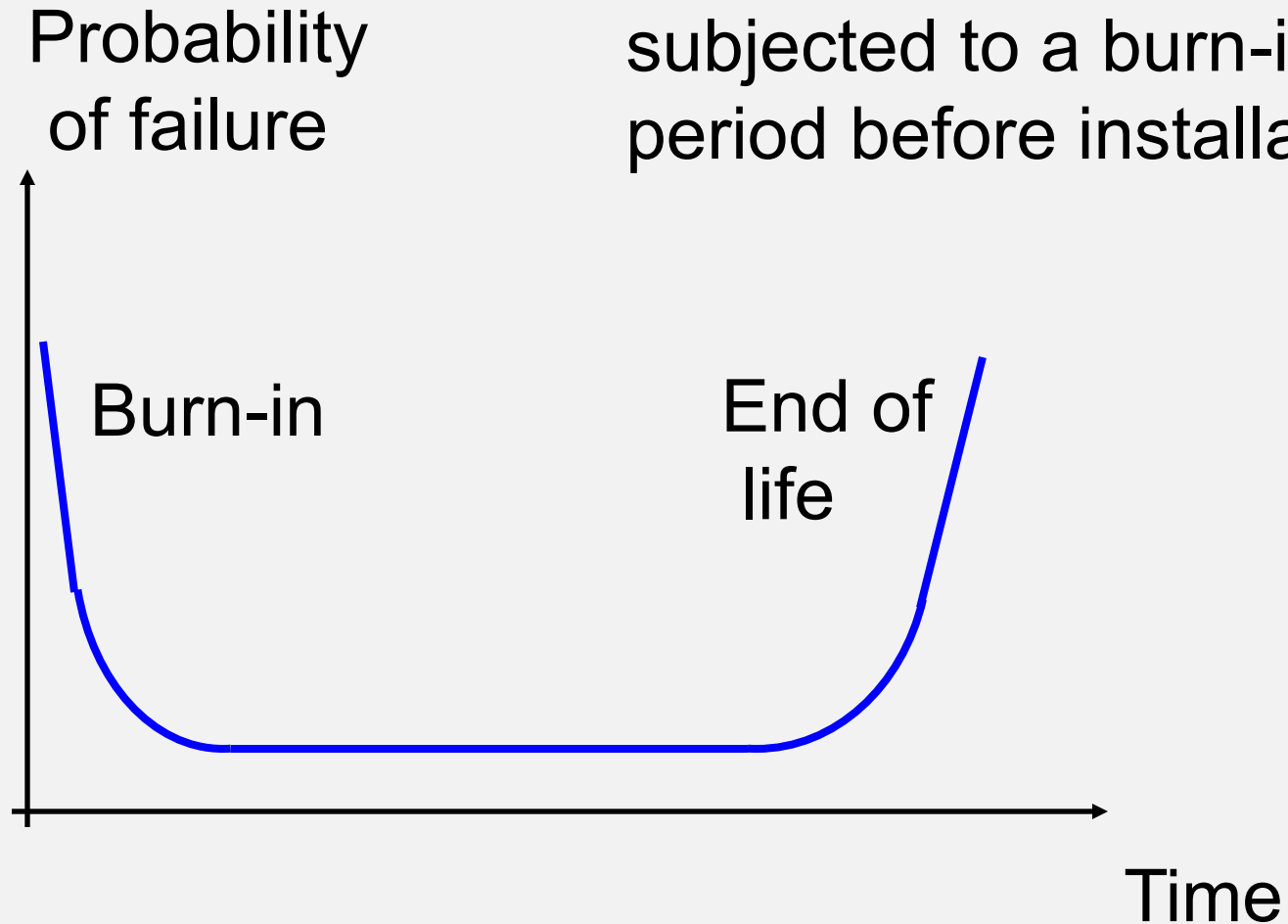


Fig 3.17 - text

Bathtub curve for reliability

Space Qualification

- Reliability can be improved by *space qualification*
- Space qualification involves:
 - selection of high reliability components
 - rigorous testing of all parts and assemblies
 - *shake and bake* tests
 - *burn in* to weed out early failures

Space Qualification

- A satellite will experience temperature extremes ($-100\text{ C} > 100\text{ C}$)
- Thermal control system is needed to manage heat flows
- A satellite will experience vibrational stress through launch
- Thermal, vacuum, and vibration testing is performed on the entire satellite in a thermal vacuum chamber in the shake-and-bake test

Summary

- Satellite has many subsystems
- Power, antennas, and communications are important for this course
- Transponders handle traffic and earn revenue
- Antennas must provide correct footprints
- High reliability is essential
- Critical components are supported with redundancy